Does Increasing Government Spending on Tertiary Education Improve the Future Tax Take? A Bayesian Causal Inference Analysis

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Abstract

When a national government increases public spending on tertiary (university) education, does that investment strengthen the country's future tax take once general economic growth is separated out? Cross-country data make this question difficult to answer because faster-growing economies naturally collect more tax revenue, international indicators divide per-student spending by gross domestic product (GDP) per person and student headcount, and university cohorts take several years to graduate and enter higher tax brackets. We assemble a harmonised five-year panel covering 121 countries from 1985 to 2024 and fit continuous-treatment Multivariate Bayesian Causal Forests that model four interrelated tax outcomes simultaneously whilst adjusting for starting country conditions, expected funding intensity, and follow-up macroeconomic growth. We find that each +10 percentage-point (pp) increase in per-student university spending as a share of GDP per person shifts the tax mix toward direct income, profit, and capital gains taxation by +0.36 pp ($P = 0.93$, where P is the posterior probability of an effect greater than zero given the data) after 5 years and +0.71 pp ($P = 0.96$) after 10 years. Overall tax revenue as a share of GDP takes a full decade to respond as university cohorts graduate and advance into higher-salaried careers, with 10-year total non-resource tax revenue rising by +0.24 pp of GDP ($P = 0.89$) and direct income tax revenue rising by +0.12 pp of GDP ($P = 0.90$) under our primary per-student ratio scaling, by +0.37 pp of GDP ($P = 1.00$) following within-country five-year funding increases, and by +0.15 pp of GDP ($P = 0.98$) per +25% increase in real purchasing power parity dollars per student. By contrast, expanding the overall university budget without protecting spending per student fails to raise tax revenue as a share of GDP, and out-of-sample validation across unseen countries confirms that our continuous-dose model generalises reliably without overfitting.

Keywords: Multivariate Bayesian Causal Forests; continuous treatment; tertiary education spending; tax capacity; direct income taxation; cross-country panel data.

1 Introduction

Public debates over higher education finance frequently centre on a simple, practical question: is government spending on tertiary (university) education a net drain on the public purse, or is it a productive investment that expands a country's future tax base? At the individual level, decades of labour economics research have shown that university graduates earn higher wages, experience lower unemployment, and work in larger, more formalised firms than non-graduates (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018; Deming, 2022). Empirical welfare analyses in the United States also indicate that direct investments in college access and per-student instructional resources can recoup a substantial share of their upfront public cost over the lifecycle through higher future income tax payments (Trostel, 2010; Chakrabarti et al., 2020; Hendren and Sprung-Keyser, 2020). As university-educated cohorts enter the labour market and raise firm productivity (Moretti, 2004; Valero

and Van Reenen, 2019), those private wage gains and corporate spillovers should in principle show up at the national level as stronger personal income, payroll, and corporate profit tax receipts.

Yet working out whether, and over what time horizon, national government spending on tertiary education actually improves a country's tax take is surprisingly tricky. First, if a country's tax revenue rises over the decade following an increase in university funding, we need to know whether universities specifically strengthened the formal tax base or whether the whole economy simply grew richer. Worse still, if an analyst constructs a national tax-per-person variable by multiplying the tax-to-GDP ratio by GDP per person, any positive effect of education on GDP automatically inflates the tax outcome by mathematical identity; isolating a genuine fiscal return requires modelling pure tax shares and tax-to-GDP ratios whilst controlling for both starting economic conditions and follow-up economic growth (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016). Second, the standard international indicator of tertiary funding intensity compiled by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) Institute for Statistics[1] and the World Bank[2] measures public expenditure per university student as a percentage of GDP per person. Because this ratio places both student headcount and GDP per person in the denominator, a low-income country with a tiny, elite university sector (such as Malawi or Ethiopia, enrolling fewer than 2% of young adults) can appear to spend over 150% of GDP per person per student, whereas a high-income country enrolling 60% of its young adults (such as Denmark or South Korea) appears to spend 27%, even though the high-income country spends nearly three times as many real purchasing power parity (PPP) dollars per student (where PPP adjusts US dollars for differences in local price levels across countries).

A third obstacle is that university investments are continuous policy choices that take time to mature and affect different parts of the tax system unequally. Treating government university spending as a simple binary on/off switch would throw away variation in funding intensity and imply that a tiny spending nudge across an arbitrary cutoff produces the same tax return as a major fiscal expansion. Moreover, a funding increase in a given five-year period immediately pays formal university salaries and supports student cohorts who take three to four years to graduate and another five years to enter peak tax-paying brackets, shifting economies away from indirect consumption and customs taxes toward direct personal income and corporate profit taxes (Gordon and Li, 2009; Besley and Persson, 2014; Kleven et al., 2016; Jensen, 2022). Figure 1 summarises our continuous-treatment study design for addressing all three obstacles across 121 countries from 1985 to 2024.

As illustrated in Figure 1, we jointly model four percentage-point (pp) tax outcomes over both +5 and +10 years: (1) total tax revenue as a share of GDP (excluding oil and mineral royalties), (2) direct income, profit, and capital gains tax revenue as a share of GDP, (3) direct income and profit taxes as a share of total tax revenue, and (4) indirect goods and services (sales and value-added) taxes as a share of total tax revenue. Rather than relying on a single spending definition, we compare four continuous measures of university funding side by side: our primary per-student spending ratio (scaled per +10 pp of GDP per person per student within each World Bank income group), within-country five-year expansions in per-student funding (per +10 pp increase), real 2021 PPP dollars per university student (per +25% relative increase, which removes GDP from the denominator), and the public university budget as a percentage of GDP (per +0.25 pp of GDP, which removes student headcount from the denominator). We also evaluate an alternative relative logarithmic scaling of the per-student spending ratio (per +25% increase) and present a full head-to-head comparison of the linear and logarithmic scalings in Appendix D. Throughout the paper, we use pp to denote percentage points, CI to denote a Bayesian posterior credible interval, and $P = P(\tau > 0 \mid \text{data})$ to denote the posterior probability that the continuous treatment slope τ is greater than zero given the observed data.

We believe this is the first global study to use modern causal inference techniques to test whether public university funding increases future tax revenue. The remainder of the paper is organised as follows. Section 2 reviews the literature on tax capacity, university resources, and Bayesian regression tree models for continuous treatments. Section 3 describes the 40-year panel dataset, the country-blocked holdout design, the UNESCO denominator trap, and where countries such as Ireland sit on the continuous funding

[1] UNESCO Institute for Statistics (UIS) education finance database: https://data.uis.unesco.org/.

[2] World Bank World Development Indicators (WDI) portal: https://datatopics.worldbank.org/world-development-indicators/.

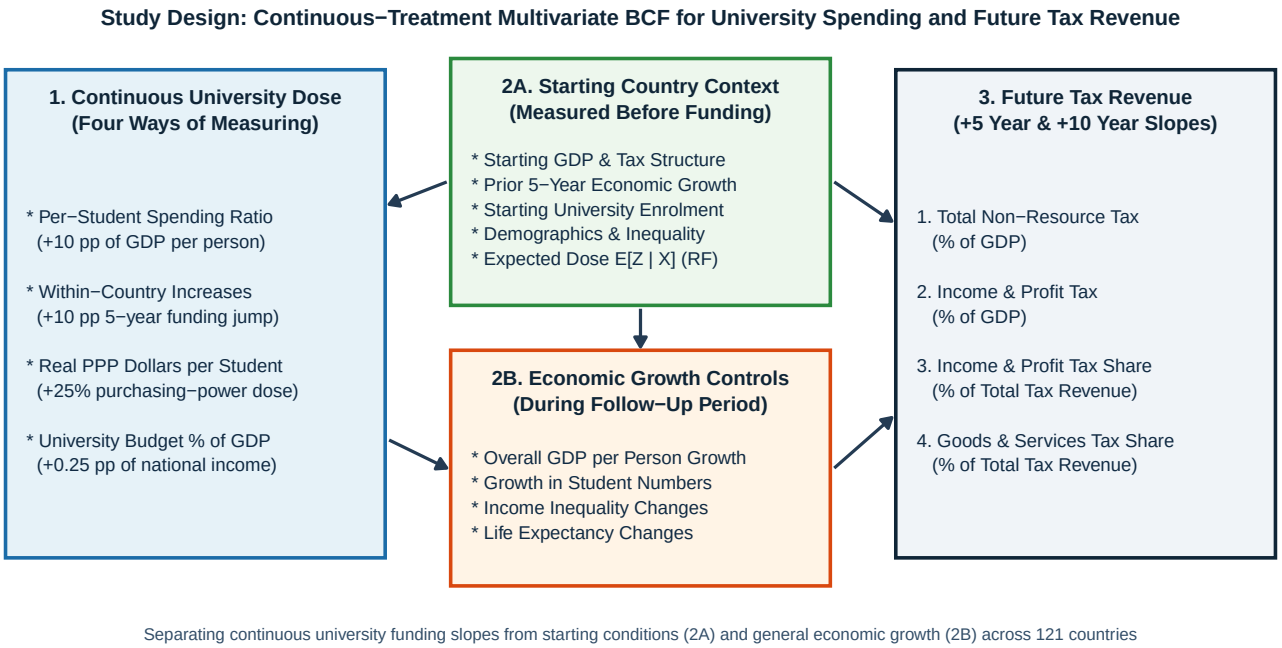


Figure 1: Overview of the study design showing the four continuous ways we measure government university funding, how we control for starting country context and expected funding dose (Box 2A) alongside follow-up economic growth (Box 2B), and the four tax revenue outcomes we track over +5 and +10 years.

scale. Section 4 explains the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) model in plain language, with full mathematical derivations in Appendix A. Section 5 presents the empirical results across time horizons, funding measures, enrolment levels, country trajectories, and out-of-sample holdout validation, whilst Section 6 discusses the policy implications, limitations, and directions for future research.

2 Literature Review

In this section we review three strands of research that have largely developed in parallel: the political economy of tax structure modernisation, the economics of per-student university resources, and Bayesian tree ensembles for causal inference. Bringing these three literatures together clarifies why university resources should affect direct income taxation more strongly than consumption taxation, and why flexible nonparametric models are needed to separate continuous funding slopes from general economic growth. Those interested in more comprehensive surveys of each field may refer to Besley and Persson (2014), Deming (2022), and Hill et al. (2020).

2.1 Tax Capacity, Formal Employment, and Tax Structure

A central question in public finance is why high-income countries collect 30% to 45% of GDP in tax revenue (primarily through direct personal income taxes, social security contributions, and corporate profit taxes), whereas low- and middle-income countries often collect less than 15% of GDP and rely heavily on indirect consumption taxes and customs duties (Besley and Persson, 2009; Gordon and Li, 2009; Besley and Persson, 2014). Early cross-country studies emphasised political institutions, property rights, and administrative state capacity (Besley and Persson, 2009; Kaufmann et al., 2011), alongside improved international accounting standards designed to separate domestic tax effort from volatile oil and mineral royalties (Prichard, 2016; UNU-WIDER, 2023).[3] Whilst these institutional studies establish that fiscal capacity evolves alongside broader economic development, they leave open the

[3]See the UNU-WIDER Government Revenue Dataset (GRD) documentation at https://doi.org/10.35188/UNU-WIDER/GRD-2023.

specific microeconomic channels through which public investments in skilled human capital expand the collectible tax base.

More recent empirical work has focussed on the microeconomic mechanics of tax enforcement, paper trails, and formal employment (Kleven et al., 2016; Keen and Slemrod, 2017; Pomeranz and Vila-Belda, 2019). Kleven et al. (2016) show that modern governments can enforce high direct tax compliance at low audit cost because large, formal firms act as third-party tax collectors: complex businesses require internal accounting records shared across managers and skilled staff, making collusive tax evasion between employers and employees unsustainable. Building on this mechanism, Jensen (2022) analyses roughly 100 household surveys across development levels and shows that as the share of salaried employees rises relative to self-employed workers, governments lower personal income tax exemption thresholds and expand direct income tax collection. Field studies confirm that electronic tax filing and third-party information trails reduce evasion when paired with capable firms and literate taxpayers (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022), whilst broader direct taxation can also strengthen citizen engagement with the state (Weigel, 2020). By contrast, in developing economies with large informal sectors, indirect consumption taxes such as value-added tax (VAT) remain the primary revenue instrument because formal retail chains only gradually replace informal vendors as incomes rise (Bachas et al., 2024).

This public finance literature suggests a direct connection between university funding and the national tax base that goes beyond headline GDP growth. Universities are themselves large formal employers whose academic and administrative staff pay income tax through automatic payroll withholding, and they train the professional, managerial, scientific, and accounting graduates required by formal corporate employers (Kleven et al., 2016; Valero and Van Reenen, 2019), shifting workers into salaried jobs above national income tax exemption thresholds (Jensen, 2022). When this mechanism operates at the national level, higher public spending per university student should specifically expand direct income, profit, and capital gains taxation as a share of total tax revenue and as a share of GDP, whilst reducing relative reliance on indirect goods and services taxes.

2.2 Returns to University Resources and Fiscal Externalities

A large microeconometric literature documents strong private wage returns to university education across both advanced and developing economies (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018; Chetty et al., 2020; Mountjoy, 2022). Reviewing this evidence, Deming (2022) notes that human capital accounts for at least one-third of within-country wage variation and at least half of cross-country income differences, with cognitive and problem-solving skills playing a key role in long-run growth (Hanushek and Woessmann, 2012; Angrist et al., 2021), even after accounting for skilled emigration from smaller developing countries (Beine et al., 2008). Beyond individual earnings, university graduates generate productivity spillovers for surrounding workers and firms: Moretti (2004) finds that a higher share of college graduates in a city raises the wages of both graduates and non-graduates, whilst Valero and Van Reenen (2019) examine roughly 15,000 universities across 1,500 sub-national regions in 78 countries and find that university expansions raise future regional GDP per person through skilled labour supply and local innovation.

Importantly for our study, higher education research demonstrates that graduation rates and subsequent earnings depend heavily on per-student instructional resources rather than enrolment headcounts alone. Bound et al. (2010) and Deming and Walters (2017) show that when public colleges expand student numbers whilst state funding per student stagnates or declines (a phenomenon known as institutional crowding), degree completion rates fall and time to graduation lengthens. Conversely, Chakrabarti et al. (2020) find that increases in state appropriations to public universities raise bachelor's degree completion and improve students' long-term financial outcomes and neighbourhood income into their thirties. When graduates complete their degrees and enter formal careers, the fiscal return to the public purse can be substantial: Trostel (2010) estimates that lifetime additional income and payroll taxes paid by college graduates exceed direct government subsidies, and Hendren and Sprung-Keyser (2020) show within their Marginal Value of Public Funds framework that direct investments in college access and per-student spending generate positive fiscal externalities through future tax receipts (though they

also find that untargeted higher education tax credits for adults yield lower welfare returns than early childhood investments).

2.3 Bayesian Regression Tree Ensembles for Continuous Treatments

Separating the fiscal effect of continuous university funding from general economic development involves two statistical challenges: controlling for intermediate economic growth without introducing bias, and accounting for non-linear confounding across diverse countries. In cross-country panels, standard linear two-way fixed effects regressions can suffer from negative weighting and functional-form misspecification when continuous treatment slopes vary across countries and over time (Jordà, 2005; Acemoglu et al., 2019; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021). Furthermore, as shown in the causal mediation literature (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016; Ting and Linero, 2025), adjusting for follow-up macroeconomic changes (such as GDP per person growth, enrolment growth, inequality, and life expectancy) allows us to isolate the direct structural effect of university spending net of overall economic growth, especially when checked via two-stage sequential g-estimation (Acharya et al., 2016).

Bayesian Additive Regression Trees (BART) (Chipman et al., 2010; Hill, 2011; Hill et al., 2020) and Bayesian Causal Forests (BCF) (Hahn et al., 2020) provide a flexible nonparametric alternative to linear panel regressions, sitting alongside frequentist causal forests and double machine learning (Chernozhukov et al., 2018; Athey et al., 2019). Tree ensembles automatically capture non-linear relationships and multi-way interactions amongst covariates, with extensions supporting high-dimensional predictors (Linero, 2018), functional smoothing (Starling et al., 2020), count outcomes (Murray, 2021), fast stochastic tree search (Krantsevich et al., 2023), causal mediation (Ting and Linero, 2025), and low-dimensional posterior summaries (Woody et al., 2021). Hahn et al. (2020) show that standard single-forest BART models can suffer from regularisation-induced confounding when strong baseline predictors of the outcome also predict treatment assignment; BCF resolves this by splitting the model into a prognostic forest $\mu(\cdot)$ and a separately regularised treatment-effect forest $\tau(\cdot) \cdot Z_{i,t}$, with extensions to multivariate outcomes and longitudinal panels (McJames et al., 2024, 2025). Because $Z_{i,t}$ enters Equation (2) multiplicatively as a scalar multiplier on the leaf vector $\tau_{l,b} \in \mathbb{R}^4$, the MV-BCF algebra and conjugate Gaussian leaf updates extend directly to a continuous treatment dose $Z_{i,t} \in \mathbb{R}$ by replacing the binary propensity score with the expected continuous treatment $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ inside the prognostic forest $\mu(\cdot)$, mirroring continuous-treatment propensity conditioning and Robinson's partially linear residualisation in causal forests and double machine learning (Robinson, 1988; Hirano and Imbens, 2004; Chernozhukov et al., 2018; Athey et al., 2019). Readers will note that the Multivariate Bayesian Causal Forest (MV-BCF) method used in this paper (McJames et al., 2025) was co-developed by the author with Nathan McJames, Ann O'Shea, and Yong Chen Goh. Given this, we want to make it clear from the outset that this paper is not written to promote one causal estimator over all others; in Section 5.4 and Appendix B, we compare the continuous-treatment MV-BCF model directly against univariate BCFs, single-forest BART, random forests, ridge regression, linear local projections, two-way fixed effects, and double machine learning, and we discuss the limitations of the BCF approach alongside its strengths.

3 Data

3.1 Global Five-Year Panel Structure and Country-Blocked Holdout Split

Annual macroeconomic series for education spending and tax revenue contain frequent reporting gaps and short-term business-cycle fluctuations. Following standard practice in empirical macro-panel studies (Acemoglu et al., 2019), we average annual observations from the World Bank WDI, the UNESCO Institute for Statistics, the International Monetary Fund (IMF) Fiscal Monitor,[4] the Worldwide Governance Indicators (WGI), the Standardised World Income Inequality Database (SWIID version 9.6; Solt

[4]IMF Fiscal Monitor database: https://www.imf.org/en/Publications/FM.

2020),[5] and the Organisation for Economic Co-operation and Development (OECD) *Education at a Glance* financing taxonomy[6] into eight non-overlapping five-year waves spanning 40 years: `1985-1989`, `1990-1994`, `1995-1999`, `2000-2004`, `2005-2009`, `2010-2014`, `2015-2019`, and `2020-2024`. Constructing five-year waves also matches the typical duration of an undergraduate degree programme and reduces serial correlation from annual budgetary carryover.

Although the full World Bank register lists 217 sovereign territories, many small island states and conflict-affected territories do not report tertiary education expenditure or disaggregated tax revenues to UNESCO and the World Bank. In addition, raw international fiscal and education archives occasionally contain reporting anomalies where currency-unit mismatches, decimal-place errors, or offshore sovereign resource funds distort headline ratios. Screening out impossible tax-to-GDP values above 60% or below 0.5% of GDP and near-zero tertiary reporting glitches below 0.5% of GDP per person removes five raw reporting anomalies in the pipeline: Sudan in `1995-1999` (recorded at 629.2% of GDP in tax due to a currency-unit mismatch), Timor-Leste in `2010-2014` (recorded at 110.0% of non-oil GDP due to offshore petroleum fund accounting), the United Arab Emirates in `2010-2014` and `2015-2019` (recorded below 0.4% of GDP in central federal tax), and Madagascar in `2015-2019` (recorded at 0.036% of GDP per person per tertiary student due to a decimal-place glitch). Requiring strictly observed university funding and four-outcome tax series (with zero imputation on treatments or outcomes) yields our analysis dataset of 121 countries: 371 five-year country-waves across 121 countries at the +5-year horizon ($t \to t+1$) and 283 country-waves across 112 countries at the +10-year horizon ($t \to t+2$).

To evaluate how well our continuous-treatment models generalise to unseen countries, we partition the 121 countries using a deterministic 5-fold country-blocked split stratified by World Bank income group. Every wave of a given country is assigned to the same fold so that no country ever straddles the training and test sets. Folds 1 to 4 (80% of countries: 98 countries and 301 country-waves at +5 years; 90 countries and 229 country-waves at +10 years) form the training sample used to compute income-group funding medians and covariate centring statistics without any test-set leakage, whilst Fold 5 (20% of countries: 23 unseen countries and 70 country-waves at +5 years; 22 unseen countries and 54 country-waves at +10 years) serves as a locked out-of-sample holdout test set evaluated in Section 5.4. In Sections 5.1 to 5.3, we report posterior treatment slopes from the full 5-fold country-blocked cross-fitted panel ($N = 371$ at +5 years and $N = 283$ at +10 years, where $\hat{z}(X_{i,t})$ is predicted strictly out-of-fold across the five country folds), whilst in Section 5.4 and Appendix Table 6 (Check `R13`) we fit the MV-BCF model exclusively on the 80% training countries in Folds 1 to 4 ($N = 229$ at +10 years, which yields even stronger level slopes of +0.42 pp on total tax/GDP, $P = 0.99$, and +0.22 pp on direct income tax/GDP, $P = 0.98$) to generate strictly out-of-sample predictions on the 22 locked holdout countries in Fold 5.

3.2 Tax Revenue Outcomes

To examine how university funding affects both overall tax collection and the mix of taxes without introducing mechanical GDP product identities, we track four percentage-point changes over +5 years ($t+1$) and +10 years ($t+2$). First, we measure total tax revenue as a percentage of GDP (ΔTax/GDP, in pp of GDP), based on the World Bank WDI tax-to-GDP series (`GC.TAX.TOTL.GD.ZS`). Following the fiscal-capacity literature (Prichard, 2016; UNU-WIDER, 2023), in major oil- and mineral-exporting countries where natural resource rents exceed 10% of GDP, we subtract half of the excess resource rents so that volatile commodity royalties do not masquerade as domestic tax effort (and we verify in Appendix B that using raw, unadjusted WDI tax-to-GDP gives very similar results).

Second, we measure direct income, profit, and capital gains tax revenue as a percentage of GDP (ΔIncomeTax/GDP, in pp of GDP), which captures revenue collected directly from individual wages, salaries, and corporate profits. Third, we track direct income, profit, and capital gains taxes as a percentage of total revenue (ΔIncomeTax/Rev, in pp of revenue), which measures structural modernisation toward progressive direct taxation (Besley and Persson, 2014; Jensen, 2022). Fourth, we track indirect goods and services taxes (such as VAT, sales taxes, and excise duties) as a percentage of total revenue

[5] SWIID version 9.6 database and documentation: `https://fsolt.org/swiid/`.

[6] OECD *Education at a Glance* portal: `https://www.oecd.org/en/publications/education-at-a-glance.html`.

(ΔGoodsTax/Rev, in pp of revenue; Bachas et al. 2024). Readers should note that Outcome 2 (direct income tax as a share of GDP) is simply the product of Outcome 1 (total tax as a share of GDP) and Outcome 3 (direct income tax as a share of revenue) divided by 100. Thus, Outcomes 1, 3, and 4 capture the two underlying policy dimensions (whether overall tax collection rises and whether the tax mix shifts from consumption taxes toward income taxes), whilst Outcome 2 measures their combined impact on the national budget. Because these four measures are algebraically linked and move together within each country's budget, a major advantage of our Multivariate Bayesian Causal Forest approach is that it models all four outcomes simultaneously through a joint covariance structure, capturing their cross-outcome dependencies directly unlike standard single-equation regressions.

3.3 The UNESCO Denominator Trap and Continuous University Funding Dose

The standard international indicator of per-student university spending compiled by UNESCO and the World Bank divides public spending per university student by GDP per person, which is algebraically identical to dividing the public university budget (% of GDP) by the number of university students per person:

$$\text{UNESCO Ratio}_{i,t} = \frac{\text{University Spend per Student}_{i,t}}{\text{GDP per Person}_{i,t}} \times 100 = \frac{\text{University Budget (\% GDP)}_{i,t}}{\text{Students per Person}_{i,t}}. \quad (1)$$

Because both GDP per person and student headcount appear in the denominator, countries with very different university systems can record similar headline ratios for completely different reasons. To see the raw data before any statistical modelling, Figure 2 plots the observed distributions of the UNESCO per-student ratio, real 2021 PPP dollars per student, gross tertiary enrolment, and the direct income tax share across the four World Bank income groups, and Table 1 reports the corresponding group summaries.

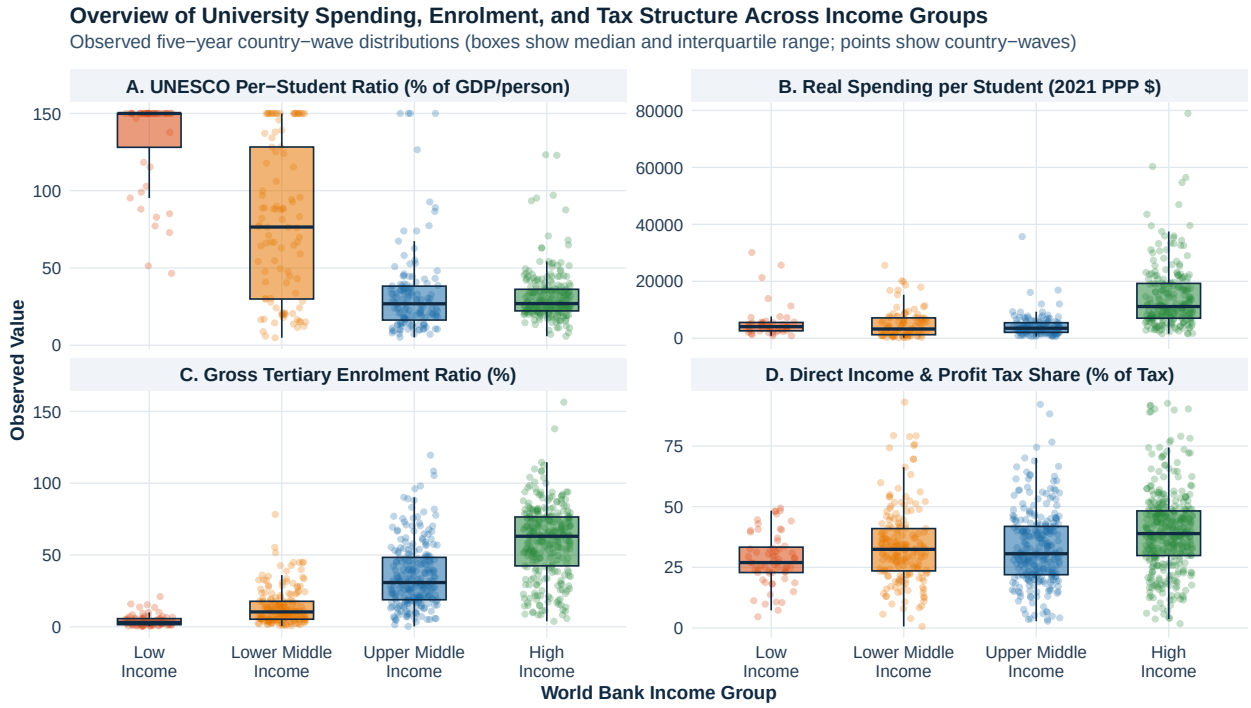


Figure 2: Observed distributions of (A) the UNESCO per-student spending ratio (% of GDP per person, winsorised at 150% for display), (B) real spending per university student in 2021 PPP dollars, (C) gross tertiary enrolment ratio (%), and (D) direct income and profit tax share (% of revenue) across the four World Bank income groups.

Figure 2 and Table 1 show clearly how the UNESCO denominator trap works in practice. In low-income countries (such as Uganda, Malawi, or Ethiopia), only 2.3% of young adults attend university; dividing a small university budget (0.47% of GDP) by a tiny student headcount and a low GDP per person

Table 1: University spending indicators, enrolment, and baseline direct income tax share across World Bank income groups. Values in brackets in the first column show the number of countries and the number of observed five-year country-waves; group medians are reported for skewed spending and enrolment variables and group means for the direct income tax share.

Income Group (Countries, Waves)	UNESCO Ratio (%)	Real PPP $ / Student	Univ. Budget (% GDP)	Tertiary Enrol. (%)	Direct Income Tax Share (%)
Low income (21, 80)	**205%**	$3,957	0.47%	2.3%	29.2%
Lower middle (41, 172)	76.5%	$3,258	0.65%	7.4%	30.8%
Upper middle (54, 219)	26.8%	$3,521	0.64%	25.5%	33.1%
High income (69, 337)	26.9%	**$11,148**	**0.94%**	**58.3%**	**38.4%**

produces a raw median UNESCO ratio of 205% (which we cap at 150% in Figure 2A and when fitting our regression trees so that extreme outliers do not distort the tree splits). That tiny elite student body also explains why median real PPP spending per student in low-income countries ($3,957, concentrated in one or two national universities) is slightly higher than in lower-middle-income countries ($3,258), where enrolment is over three times higher (7.4% median) and public budgets are stretched across many more students. Once we unbundle the denominators in Table 1, high-income countries (such as Denmark, Norway, or South Korea) spend nearly three times as much per student in real purchasing power ($11,148 vs $3,957) and twice as much of their national income (0.94% vs 0.47% of GDP), whilst collecting a larger share of their tax revenue from direct income and profit taxes (38.4% vs 29.2%).

Because comparing raw UNESCO ratios across income groups would conflate small elite enrolment with high per-student funding, we construct our primary continuous university funding dose ($Z_{i,t}$, which we refer to as Scaling A) by centring each country-wave's winsorised per-student spending ratio on the training-set median of its own World Bank income group ($m_{g(i,t)} = 150.0\%$ for low income after the 150% pre-winsorisation cap, vs 183.8% raw before winsorisation, 68.9% for lower-middle income, 27.6% for upper-middle income, and 26.4% for high income), dividing by 10 percentage points, winsorising at ± 1.5 dose units (± 15 pp around the group median) so that extreme archival ratios do not exert undue leverage, and adding $+1.5$ so that $Z_{i,t} \in [0, 3.0]$ is strictly non-negative with $Z_{i,t} = 1.5$ representing the income-group median and a unit increase ($\Delta Z_{i,t} = +1.0$, $\mathrm{SD}(Z) = 1.05$) representing a $+10$ pp increase in per-student funding as a share of GDP per person. On this continuous scale, Nordic and continental European high-income countries with generous public university funding sit near the top of the distribution (such as Norway and Sweden at $Z_{i,t} = 2.7$ to 3.0, Denmark at 3.0, and Austria at 2.4 to 2.5), countries sitting near their income-group median record $Z_{i,t} \approx 1.5$ (such as New Zealand and Iceland at 1.47), and countries relying more heavily on private tuition or lower public allocations sit below 1.5 (such as Spain at 1.10, the United States at 0.91, Chile at 0.81, and South Korea at 0.35). Ireland illustrates a striking within-country trajectory along this continuous dose: Ireland sat above the high-income median in `1995–1999` (28.2%, $Z_{i,t} = 1.68$), `2005–2009` (28.4%, $Z_{i,t} = 1.70$), and `2010–2014` (27.6%, $Z_{i,t} = 1.62$), dipped slightly below it in `2000–2004` (25.6%, $Z_{i,t} = 1.42$), and dropped to 17.3% ($Z_{i,t} = 0.59$, or 9.1 pp below the median) in `2015–2019` as student numbers grew much faster than state grants after the 2008 financial crisis. In Appendix D, we compare this primary linear $+10$ pp ratio dose (Scaling A) against a relative logarithmic dose (Scaling B, which centres the raw log series on the unclipped training medians of 183.8%, 68.9%, 27.6%, and 26.4% and therefore preserves low-income variation above 150%) and show that both scalings yield closely aligned substantive conclusions and out-of-sample predictive performance.

4 Methodology

In an ideal experiment, we would randomly assign different countries to increase per-student university funding by varying amounts and then track their tax receipts over the next decade. In observational data, however, governments set their university budgets in response to economic conditions, demographics, and existing tax capacity, and the relationships among these variables are strongly non-linear (Hill,

2011; Hill et al., 2020). To separate the causal slope of continuous university funding from confounding macroeconomic trends whilst modelling all four interrelated tax outcomes simultaneously, we use the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) model (Hahn et al., 2020; McJames et al., 2025), providing the intuitive setup here and full mathematical derivations in Appendix A.

The Multivariate Bayesian Causal Forest (Hahn et al., 2020; McJames et al., 2025) splits the prediction of future tax changes into two distinct regression tree ensembles (Chipman et al., 2010): a baseline prognostic trajectory (what would have happened to a country's tax take naturally given its starting conditions, expected funding level, and overall economic growth) and a continuous causal treatment slope that can vary with country characteristics. For country i at wave t and follow-up horizon $h \in \{1, 2\}$ (+5 and +10 years), the four tax changes $\Delta Y_{i,t+h} \in \mathbb{R}^4$ are modelled jointly as:

$$\Delta Y_{i,t+h} = \underbrace{\mu(X_{i,t}, \hat{z}(X_{i,t}))}_{\text{Baseline Trajectory}} + \underbrace{\tau(W_{i,t})}_{\text{Causal Slope}} \cdot Z_{i,t} + \varepsilon_{i,t+h}, \quad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma). \tag{2}$$

To estimate the baseline trajectory $\mu(X_{i,t}, \hat{z}(X_{i,t}))$, we use an ensemble of 75 shallow decision trees (many having only one or two splits), where each terminal leaf outputs a vector of four values corresponding to our four tax outcomes. The baseline forest is fed with 19 pre-treatment country characteristics measured at or before wave t (starting log real GDP per person in PPP dollars, starting total tax-to-GDP ratio, starting direct income tax share, starting goods and services tax share, disposable income Gini inequality, life expectancy, gross tertiary and secondary enrolment ratios, total education spending as a share of GDP, working-age population share, log population, unemployment rate, natural resource rents, trade openness, lagged five-year GDP per person growth, World Bank income group, geographic region, OECD membership, and wave time index) together with four follow-up macroeconomic controls over $[t, t + h]$ (forward GDP per person growth, university enrolment expansion, income inequality changes, and life expectancy changes) so that general economic growth during the follow-up window is separated out (Imai et al., 2010; Acharya et al., 2016). Because treatment $Z_{i,t} \in [0, 3.0]$ is continuous rather than binary, the binary propensity score $P(Z = 1 \mid X)$ no longer applies; instead, we estimate the expected continuous funding dose $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ from the 19 pre-treatment covariates using a 500-tree random forest (`ranger`; Breiman 2001; Wright and Ziegler 2017) with country-blocked cross-fitting and include $\hat{z}(X_{i,t})$ as a prognostic covariate inside $\mu(\cdot)$. Including $\hat{z}(X_{i,t})$ guards against regularisation-induced confounding (Hahn et al., 2020) and allows the prognostic forest to absorb baseline selection into higher funding doses; in Appendix C, we show that residualising $Z_{i,t}$ on $\hat{z}(X_{i,t})$ reduces the mean absolute correlation between treatment and the 19 pre-treatment covariates by more than 60% (from 0.157 to 0.061).

To allow the fiscal slope per +10 pp of university funding to vary across countries without chasing noise in a macro panel, we model the treatment slope $\tau(W_{i,t})$ using a second, smaller ensemble of 30 shallower trees fed with 8 country moderators $W_{i,t}$ (gross tertiary enrolment, log real GDP per person, gross secondary enrolment, starting direct income tax share, working-age population share, World Bank income group, OECD membership, and wave index). Using a larger, more flexible tree prior for the prognostic forest $\mu(\cdot)$ and a smaller, more conservative tree prior for the treatment-effect forest $\tau(\cdot)$ is standard practice in Bayesian causal forest modelling (following Chipman et al. 2010; Hill 2011; Hahn et al. 2020), as it allows the model to absorb complex baseline confounding whilst shrinking noisy subgroup slope differences toward the overall average slope. All four tax outcomes are linked through the 4×4 residual covariance matrix Σ, which receives a conjugate Inverse-Wishart prior (with all prior hyperparameters detailed in Appendix A).

The main advantage of the continuous-treatment MV-BCF framework is that it estimates smooth dose-response slopes, captures non-linear confounding, and shares information across correlated tax outcomes whilst providing full Bayesian posterior distributions for every country and horizon. The main disadvantage is that tree ensembles rely on Markov chain Monte Carlo (MCMC) sampling and conditional ignorability assumptions rather than automatic linear fixed-effect differencing, and assume locally linear dose slopes within terminal treatment leaves over the winsorised range $Z_{i,t} \in [0, 3.0]$. All MV-BCF models are fitted in C++ for 2,000 MCMC iterations (750 burn-in, 1,250 retained posterior draws), with convergence, residual, and holdout validation diagnostics reported in Section 5.4 and Appendix C.

5 Results

5.1 Continuous Funding Slopes After 5 and 10 Years

We begin by examining our primary question: when a country increases its per-student university spending ratio by +10 percentage points of GDP per person ($\Delta Z = +1.0$ unit on our continuous Scaling A dose), what happens to its tax revenue 5 and 10 years later once starting country conditions, expected funding dose, and follow-up economic growth are controlled for? Recall that throughout all tables and figures, CI denotes a Bayesian posterior credible interval and $P = P(\tau > 0 \mid \text{data})$ denotes the posterior probability that the continuous treatment slope is greater than zero given the data. Table 2 and Figure 3 report the posterior mean treatment slopes, 80% and 95% credible intervals, and posterior probabilities P across the four tax outcomes at +5 years ($N = 371$, 121 countries) and +10 years ($N = 283$, 112 countries).

Table 2: Posterior mean continuous treatment slopes (in percentage points, pp, per +10 pp increase in per-student university spending as a share of GDP per person), 80% and 95% credible intervals (CIs), and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ from the four-outcome Multivariate Bayesian Causal Forest at +5 years ($N = 371$, 121 countries) and +10 years ($N = 283$, 112 countries).

Tax Revenue Outcome (Scaling A)	+5 Years Mean (P)	+5 Years 80% CI / 95% CI	+10 Years Mean (P)	+10 Years 80% CI / 95% CI
1. Total Non-Resource Tax / GDP (% of GDP, pp per +10 pp dose)	−0.06 ($P = 0.31$)	[−0.20, +0.09] [−0.28, +0.18]	+0.24 ($P = 0.89$)	[−0.01, +0.48] [−0.15, +0.70]
2. Direct Income & Profit Tax / GDP (% of GDP, pp per +10 pp dose)	+0.05 ($P = 0.81$)	[−0.02, +0.13] [−0.07, +0.17]	+0.12 ($P = 0.90$)	[−0.002, +0.26] [−0.08, +0.33]
3. Direct Income & Profit Tax Share (% of total tax revenue, pp per +10 pp dose)	+0.36 ($P = 0.93$)	[+0.04, +0.68] [−0.13, +0.81]	+0.71 ($P = 0.96$)	[+0.18, +1.17] [−0.08, +1.42]
4. Goods & Services Tax Share (% of total tax revenue, pp per +10 pp dose)	+0.10 ($P = 0.62$)	[−0.28, +0.49] [−0.47, +0.67]	−0.42 ($P = 0.14$)	[−0.91, +0.07] [−1.19, +0.34]

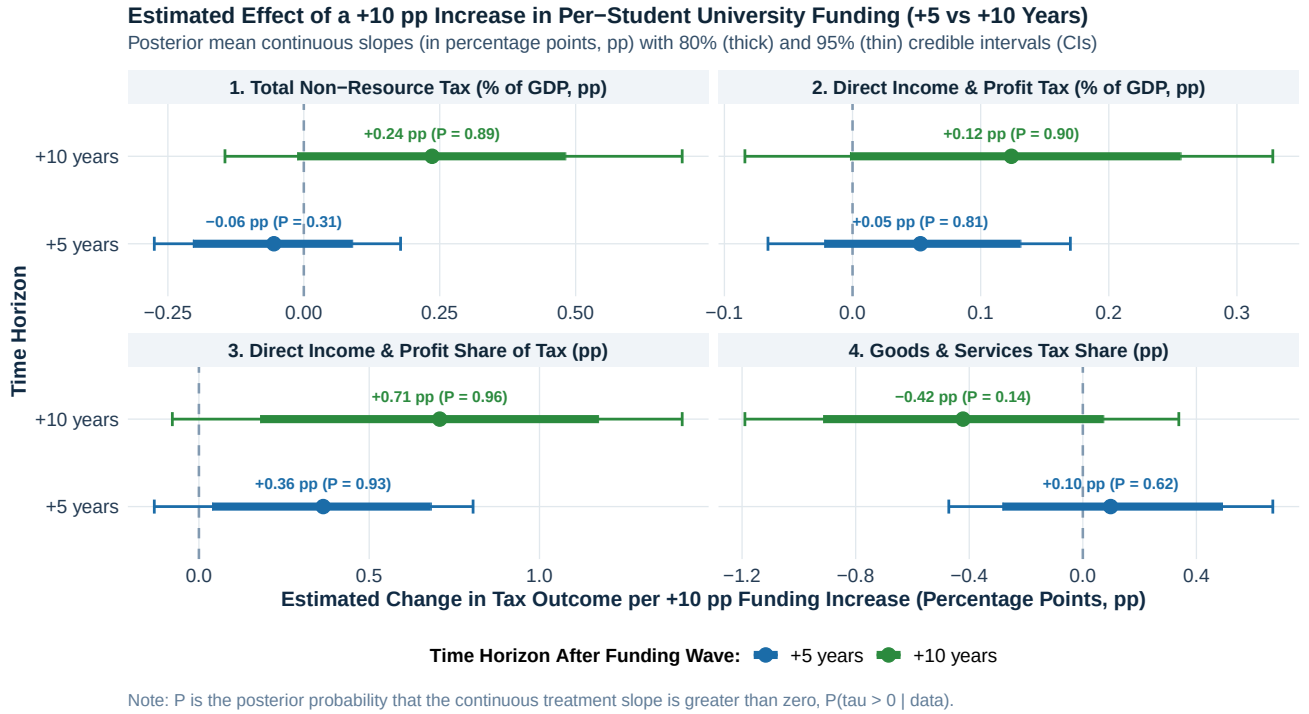


Figure 3: Posterior mean continuous treatment slopes (in percentage points, pp, per +10 pp increase in per-student university spending) with thick 80% and thin 95% credible intervals (CIs) and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ across the four tax outcomes at +5 years (blue) and +10 years (green).

Table 2 and Figure 3 show a clear maturation pattern across the two time horizons. First, higher

per-student university funding produces a progressive and lasting shift in the tax mix toward direct income, profit, and capital gains taxation: each +10 pp increase in the per-student spending ratio raises the direct income tax share of revenue (Outcome 3) by +0.36 pp after 5 years (80% CI $[+0.04, +0.68]$, 95% CI $[-0.13, +0.81]$, $P = 0.93$) and by +0.71 pp after 10 years (80% CI $[+0.18, +1.17]$, 95% CI $[-0.08, +1.42]$, $P = 0.96$). Second, overall tax collection as a share of GDP takes a full decade to respond: after 5 years, total non-resource tax revenue (Outcome 1) is slightly negative at −0.06 pp of GDP (95% CI $[-0.28, +0.18]$, $P = 0.31$) and direct income tax revenue (Outcome 2) is +0.05 pp of GDP ($P = 0.81$), whereas after 10 years, as university cohorts complete their degrees and advance into higher-salaried careers, total non-resource tax revenue rises by +0.24 pp of GDP per +10 pp funding dose (80% CI $[-0.01, +0.48]$, 95% CI $[-0.15, +0.70]$, $P = 0.89$) and direct income tax revenue rises by +0.12 pp of GDP (80% CI $[-0.002, +0.26]$, 95% CI $[-0.08, +0.33]$, $P = 0.90$; when follow-up macroeconomic growth controls are omitted to measure the total fiscal effect in Appendix Table 5, these 10-year slopes strengthen to +0.30 pp of GDP on total non-resource tax, $P = 0.95$, and +0.16 pp of GDP on direct income tax, $P = 0.96$). Third, on the indirect goods and services tax share (Outcome 4), the 10-year slope is negative at −0.42 pp ($P = 0.14$, or −0.65 pp, $P = 0.04$ for the total effect), confirming that university investment shifts the tax structure away from indirect consumption taxes and toward direct income and profit taxes. To put these continuous slopes into concrete national currency terms (holding each country's latest population and 2021 PPP GDP per person constant so that demographic growth does not inflate the calculation), a 10-year direct income tax gain of +0.12 to +0.16 pp of GDP per +10 pp funding increase corresponds to roughly €0.23 billion to €0.30 billion per year in additional income tax receipts for the median-sized economy in our sample (total GDP of €183 billion), or around €0.36 billion to €0.68 billion per year in a medium-sized high-income economy such as Ireland and €4.0 billion to €5.2 billion per year in a large economy such as the United Kingdom (we report percentage points of GDP in cross-country charts because national GDP spans four orders of magnitude across our 121 countries).

5.2 Comparing the Four Continuous Ways of Measuring University Funding

Whilst Figure 3 compared the +5-year and +10-year time horizons for our primary per-student spending ratio, a natural follow-up question is how the +10-year fiscal slope changes when we unbundle the UNESCO ratio into its separate continuous components. In Table 3 and Figure 4, we hold the time horizon fixed at +10 years and compare four distinct continuous ways of measuring university funding: (1) our primary per-student spending ratio (per +10 pp of GDP per person per student, Scaling A), (2) observed within-country five-year changes in per-student funding (per +10 pp five-year expansion), (3) real 2021 PPP dollars per student (per +25% relative log increase, which removes GDP per person from the denominator), and (4) the total public university budget as a percentage of GDP (per +0.25 pp of GDP, which removes student headcount from the denominator). Comparing these four continuous doses on the same 10-year horizon allows us to test whether future tax gains are driven by real instructional resources per student or merely by aggregate budget size.

Table 3: Comparison of +10-year posterior mean continuous slopes (in percentage points, pp), 95% credible intervals (CIs), and posterior probabilities $P = P(\tau > 0 \mid \text{data})$ across the four continuous ways of measuring university funding. The second column lists the sample size N and number of countries in brackets.

Continuous University Funding Measure (+10-Year Horizon)	Waves (Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
1. Per-Student Ratio (+10 pp GDP/person)	283 (112)	+0.24 ($P = 0.89$) [−0.15, +0.70]	+0.12 ($P = 0.90$) [−0.08, +0.33]	+0.71 ($P = 0.96$) [−0.08, +1.42]
2. Within-Country 5y Change (+10 pp)	176 (86)	−0.29 ($P = 0.17$) [−0.91, +0.31]	+0.37 ($P = 1.00$) [+0.08, +0.65]	+1.56 ($P = 0.99$) [+0.36, +2.73]
3. Real PPP Dollars / Student (+25% log)	283 (112)	+0.22 ($P = 0.95$) [−0.04, +0.49]	+0.15 ($P = 0.98$) [+0.002, +0.30]	+0.47 ($P = 0.95$) [−0.12, +1.05]
4. University Budget % of GDP (+0.25 pp)	403 (123)	−0.06 ($P = 0.37$) [−0.40, +0.28]	−0.04 ($P = 0.34$) [−0.23, +0.13]	+0.53 ($P = 0.94$) [−0.13, +1.17]

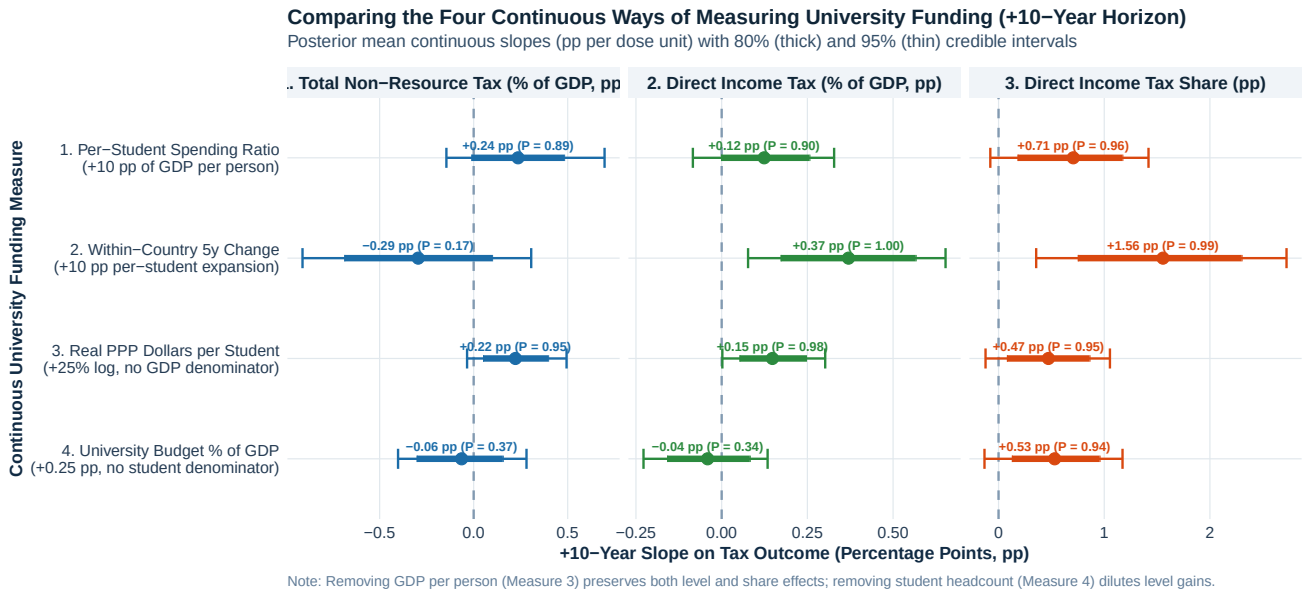


Figure 4: Comparison of +10-year posterior mean continuous slopes (in percentage points, pp, with thick 80% and thin 95% CIs) across the four continuous ways of measuring university funding.

The comparison in Table 3 and Figure 4 shows why unbundling the denominator is so informative. When we remove GDP per person from the denominator and measure continuous funding in real 2021 PPP dollars per student (Row 3, scaled per +25% relative increase around each income group's median), the +10-year posterior precision sharpens substantially: total non-resource tax revenue rises by +0.22 pp of GDP per +25% dose (80% CI $[+0.05, +0.40]$, 95% CI $[-0.04, +0.49]$, $P = 0.95$), direct income tax as a share of GDP rises by +0.15 pp (95% CI $[+0.002, +0.30]$, $P = 0.98$), and the direct income tax share rises by +0.47 pp ($P = 0.95$, following a +0.59 pp gain at +5 years with 95% CI $[+0.18, +1.00]$, $P = 1.00$). Even more notably, when we isolate dynamic within-country five-year expansions in per-student funding (Row 2, $N = 176$ across 86 countries, scaled per +10 pp five-year increase), the +10-year slope on direct income tax revenue triples to +0.37 pp of GDP (95% CI $[+0.08, +0.65]$, $P = 1.00$) and the direct income tax share jumps by +1.56 pp (95% CI $[+0.36, +2.73]$, $P = 0.99$), even as total tax-to-GDP is slightly negative (−0.29 pp, $P = 0.17$) because governments that rapidly increase per-student spending often simultaneously phase out distortionary indirect trade and excise taxes (−0.94 pp on the goods tax share, $P = 0.09$). Under the bilinear identity $Y^{(2)} = Y^{(1)} \times Y^{(3)}/100$, both effects are mutually consistent because the countries undertaking sustained within-country per-student expansions operate at higher baseline tax-to-GDP ratios, so a +1.56 pp shift in the direct income tax share raises direct income tax as a share of GDP by +0.37 pp even when indirect trade taxes are reduced. By contrast, when we remove student headcount from the denominator and look only at the aggregate university budget as a share of GDP without accounting for how many students share that budget (Row 4, per +0.25 pp of GDP), the +10-year slopes on total tax revenue (−0.06 pp, $P = 0.37$) and direct income tax revenue (−0.04 pp, $P = 0.34$) are slightly negative and indistinguishable from zero, consistent with microeconomic evidence that expanding student headcounts without protecting per-student resources leads to institutional crowding and lower completion rates (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020).

5.3 Subgroup Moderation, Negative Scenarios, and Cohort Maturation Trajectories

How do the continuous fiscal slopes vary across countries at different stages of university enrolment and economic development, and are there situations where increasing university funding reduces the tax take? Figure 5 plots the +10-year subgroup posterior mean continuous slopes and 95% credible intervals across starting university enrolment tiers and World Bank income groups for total non-resource tax/GDP, direct income tax/GDP, and the direct income tax share (with full numerical breakdowns, including our five-group OECD financing regime sub-model, reported in Appendix Table 4). As shown

in Figure 5, the two fiscal dimensions vary in contrasting directions across development stages. On the tax-mix dimension (Panel 3), the shift toward direct income and profit taxation per +10 pp of per-student funding rises as economies develop a broad formal corporate and salaried sector: it is smallest in low-income economies (+0.16 pp, $P = 0.69$), averages +0.64 pp ($P = 0.91$) in elite systems with under 20% enrolment, rises to +0.74 pp ($P = 0.96$) in lower-middle-income economies, +0.75 pp ($P = 0.97$) in massifying systems with 20–50% enrolment, and +0.84 pp (95% CI $[+0.09, +1.57]$, $P = 0.98$) in upper-middle-income economies, and remains strongly positive at +0.68 to +0.73 pp ($P = 0.94$–0.95) in high-income and high-participation systems ($\geq 50\%$ enrolment, and +1.05 to +1.14 pp, $P = 0.93$–0.95 across all five OECD financing regimes). By contrast, on the aggregate tax-to-GDP dimension (Panels 1 and 2), the largest gains per +10 pp dose occur in low-income (+0.48 pp of GDP on total tax and +0.24 pp on direct income tax, $P = 0.91$), lower-middle-income (+0.32 pp and +0.16 pp, $P = 0.91$–0.93), and elite-enrolment (+0.34 pp and +0.17 pp, $P = 0.91$–0.93) economies whose formal tax bases are still expanding, tapering off to +0.19 pp ($P = 0.82$) and +0.10 pp ($P = 0.81$) in mature high-income economies.

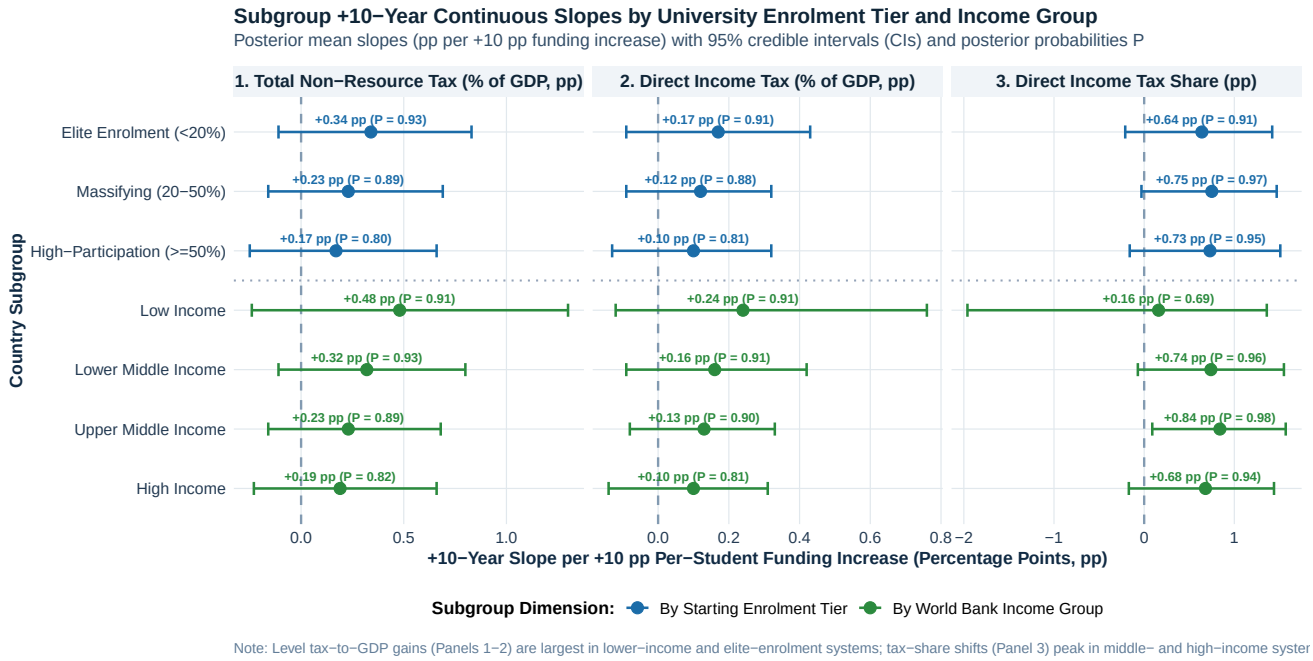


Figure 5: Subgroup +10-year posterior mean continuous slopes (in percentage points, pp per +10 pp per-student funding increase, with horizontal 95% credible interval bars and posterior probabilities $P = P(\tau > 0 \mid \text{data})$) across starting university enrolment tiers and World Bank income groups for total non-resource tax/GDP, direct income tax/GDP, and the direct income tax share.

Both the historical data and our continuous-treatment model estimates also highlight three specific scenarios where increasing university funding is associated with a *decrease* in the tax take. First, in the short run (+5 years), 243 out of 371 country-waves (65.5%) have negative point estimates on total tax-to-GDP (down to −0.96 pp of GDP per +10 pp funding increase, with an overall mean of −0.06 pp and −0.30 pp in low-income economies), reflecting a temporary "student lock-in" and cohort-gestation effect: even when follow-up enrolment headcount is controlled for, higher per-student funding supports intensive-margin study persistence and full-time degree completion so that young adults spend more time in lecture theatres rather than working full time during those initial five years. Second, at the +10-year horizon in very low-income economies with tiny elite enrolment below 2% and extreme UNESCO per-student ratios above 150% (such as Ethiopia in `1995-1999` at −0.94 pp and Malawi in `2010-2014` at −0.68 pp on the direct income tax share), pouring additional subsidies into a tiny elite university sector without broad formal-sector job creation or protection against skilled emigration (Beine et al., 2008) fails to modernise the domestic income tax share. Third, in mature OECD economies where total tax revenue already exceeds 35% of GDP (Rows 8 to 12 of Appendix Table 4, where total tax-to-GDP slopes average −0.14 to −0.17 pp per +10 pp dose), expanding public university budgets can coincide with a slightly lower overall tax-to-GDP ratio even as the income tax share of revenue rises strongly by

+1.05 to +1.14 pp ($P = 0.93$–0.95).

To illustrate how these effects evolve from the initial +5-year window to the +10-year window across development stages, Figure 6 plots the five-year and 10-year cohort maturation trajectories across the four World Bank income groups alongside the global 80% and 95% posterior credible fan. In Panel A of Figure 6, total non-resource tax revenue as a share of GDP dips below zero across every income group at the +5-year mark (−0.30 pp in low-income, −0.04 pp in lower-middle-income, −0.01 pp in upper-middle-income, and −0.06 pp in high-income economies) whilst students are still enrolled in university, before rebounding into positive territory across all four groups by the +10-year mark (+0.48 pp in low-income, +0.32 pp in lower-middle-income, +0.23 pp in upper-middle-income, and +0.19 pp in high-income economies). In Panel B of Figure 6, the direct income and profit tax share of revenue already rises at +5 years across all four groups (+0.29 to +0.49 pp) and then diverges sharply at +10 years: in lower-middle-income (+0.74 pp), upper-middle-income (+0.84 pp), and high-income (+0.68 pp) economies where formal salaried employment can absorb new graduates, the compositional shift toward direct income taxation roughly doubles by year 10, whereas in low-income economies it flattens out at +0.16 pp. In Appendix B, we also summarise our extensive sensitivity checks across competing Bayesian and frequentist estimators, within-country fixed-effect demeaning, non-overlapping decadal waves, pre-trend controls, two-stage sequential g-estimation, pre-COVID samples, the 80% training-only country sample, and the legacy binary median-cutoff specification; those checks confirm that the 10-year shift toward direct income taxation holds across every specification.

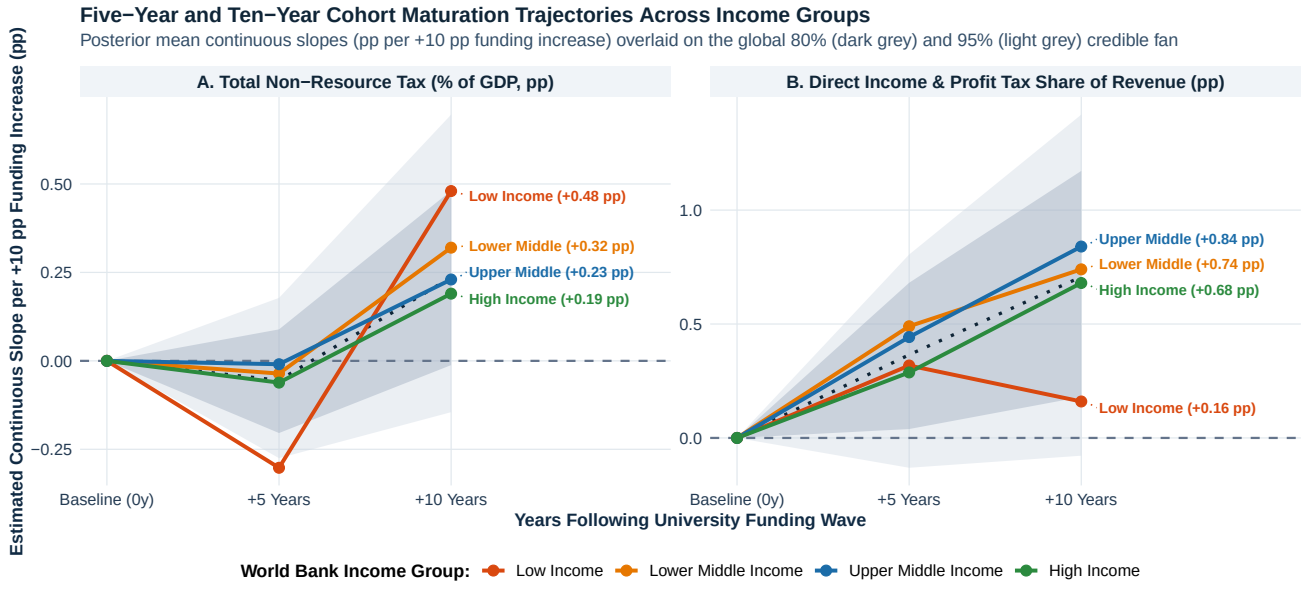


Figure 6: Five-year ($N = 371$) and 10-year ($N = 283$) horizon-specific cohort maturation trajectories of the continuous slopes per +10 pp university spending increase across World Bank income groups for (A) total non-resource tax revenue as a percentage of GDP (pp) and (B) the direct income and profit tax share of revenue (pp), overlaid on the global posterior mean (dotted line) and the 80% (dark grey) and 95% (light grey) posterior credible fan.

5.4 Model Fit, Residual Diagnostics, and Out-of-Sample Holdout Validation

Because cross-country macro panels have modest sample sizes, it is essential to check that our continuous-treatment MV-BCF model fits the observed data without systematic residual bias across the treatment range $Z_{i,t} \in [0, 3.0]$ and generalises accurately to countries that were excluded from the training run. In particular, if the linear dose multiplier $Z_{i,t}$ inside the treatment trees were misspecified or if the 75 prognostic trees were overfitting idiosyncratic country histories, we would see systematic curvature in the residuals or poor out-of-sample forecasts on unseen economies. Figure 7 presents our three core model-fit and out-of-sample validation checks at the +10-year horizon: (A) standardised posterior residuals plotted across the continuous university funding dose $Z_{i,t} \in [0, 3.0]$, (B) 5-fold country-blocked out-of-sample

cross-validation continuous ranked probability scores (CRPS, where lower values indicate more accurate probabilistic forecasts; Diebold and Mariano 1995; Gneiting and Raftery 2007) across all 112 countries ($N = 283$), and (C) out-of-sample holdout predictions vs observed 10-year changes in the direct income tax share on the 22 unseen test countries in Fold 5 ($N = 54$ country-waves).



Figure 7: Continuous-treatment model fit, residual diagnostics, and out-of-sample holdout validation at +10 years: (A) standardised posterior residuals across the continuous per-student spending dose $Z_{i,t} \in [0, 3.0]$ with LOESS smooths, (B) 5-fold country-blocked cross-validation error (CRPS) across all 112 countries comparing MV-BCF against competing models, and (C) out-of-sample predicted vs observed 10-year changes in the direct income tax share across the 22 unseen holdout countries in Fold 5.

Three diagnostic results stand out from Figure 7 (with full numerical tables reported in Appendix C and Appendix D). First, in Panel A, the LOESS trend curves across $Z_{i,t} \in [0, 3.0]$ are flat and centred on zero for all four tax outcomes, with 94.0% to 95.8% of standardised posterior residuals lying within ± 2 standard deviations, empirical 95% posterior predictive coverage between 95.8% and 97.9%, and near-zero residual skewness (-0.30 to $+0.17$), confirming that our linear +10 pp dose scaling captures the dose-response relationship without curvature or tail distortion. Second, in Panel B, across 5-fold country-blocked cross-validation on all 112 countries, our continuous MV-BCF model achieves a mean

out-of-sample CRPS of 2.33 pp (1.64 pp on total tax/GDP, 0.85 pp on direct income tax/GDP, 3.41 pp on the direct income tax share, and 3.43 pp on the goods tax share, with 81.5% empirical 80% coverage and 92.8% empirical 95% coverage), outperforming historical training means (2.45 pp), ridge regression (2.52 pp), a zero-change random walk baseline (2.53 pp; Diebold-Mariano $p = 0.062$ on the direct income tax share), and linear local projections (2.64 pp; $p = 0.053$ on the direct income tax share and $p = 0.023$ on the goods tax share) whilst matching nonparametric random forests (2.35 pp) and single-forest BART (2.33 pp). Third, in Panel C, when trained exclusively on the 90 countries in Folds 1 to 4 and evaluated on the 22 completely unseen holdout countries in Fold 5 ($N = 54$ waves), the MV-BCF model achieves out-of-sample 80% interval coverage of 75.9% to 85.2%, 95% coverage of 88.9% to 96.3%, and positive out-of-sample predicted treatment slopes across the unseen countries ($\bar{\tau}_{\text{test}} = +0.44$ pp on total tax/GDP, +0.21 pp on direct income tax/GDP, and +0.72 pp on the direct income tax share). Readers interested in a detailed comparison between our primary linear +10 pp ratio dose (Scaling A) and the relative logarithmic +25% dose (Scaling B) across all in-sample and out-of-sample diagnostics are referred to Appendix D.

6 Conclusions and Discussion

6.1 Cohort Maturation Timelines and Policy Implications

Our findings across 121 countries show that government spending on university education produces a measurable fiscal return, but that this return operates primarily by shifting the tax structure toward direct income, profit, and capital gains taxation and requires a full 10-year horizon to mature. Because our panel averages annual data into five-year waves, a continuous university funding increase in wave t (years 0 to 4) is evaluated in wave $t + 1$ (years 5 to 9) and wave $t + 2$ (years 10 to 14). During the first five-year window, a +10 pp increase in per-student university funding immediately expands formal university payrolls (whose academic and administrative staff pay income tax via automatic employer withholding) whilst keeping a larger cohort of young adults in full-time study, which shifts the tax mix toward direct income tax (+0.36 pp, $P = 0.93$) without raising overall tax-to-GDP (−0.06 pp). By the 10-year window (years 10 to 14), two university cohorts have completed their degrees and moved into salaried professional, managerial, and technical roles inside formal firms (Kleven et al., 2016; Valero and Van Reenen, 2019; Jensen, 2022), raising the direct income tax share by +0.71 pp ($P = 0.96$), direct income tax revenue by +0.12 to +0.37 pp of GDP, and total non-resource tax revenue by +0.22 to +0.30 pp of GDP.

For finance and education ministries, our comparison of the four continuous funding definitions in Section 5.2 carries a clear practical lesson: expanding the overall university budget or student numbers without protecting funding *per student* yields weak fiscal returns (−0.04 pp on direct income tax/GDP when only budget share of GDP is considered). Countries that maintain or expand real resources per student avoid institutional crowding (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020) and achieve much stronger 10-year gains in direct income tax receipts (+0.15 pp of GDP per +25% increase in real PPP dollars per student and +0.37 pp of GDP per +10 pp within-country five-year expansion). To put these magnitudes in perspective using Table 1 and Appendix Table 7, across the 44 countries sitting below their income-group median in `2015-2019`, raising per-student spending by a full +10 pp of GDP per person costs a median of 0.40% of GDP per year (0.29% to 0.81% of GDP in most high-income economies, where a smaller +2.5 to +5.0 pp boost represents 0.10% to 0.20% of GDP). In national currency terms (holding population and baseline GDP constant), for a median-sized economy of €183 billion GDP, an upfront per-student funding boost costing 0.20% to 0.40% of GDP (€365 million to €730 million per year) generates an estimated €225 million to €675 million per year in additional direct income tax revenue after 10 years when only two five-year graduate cohorts have entered their late twenties and early thirties (whereas a full 40-year career steady state spans seven to eight active five-year cohorts paying higher direct taxes simultaneously, consistent with microeconomic lifecycle analyses; Trostel 2010; Hendren and Sprung-Keyser 2020).

6.2 Limitations and Future Research

Several caveats apply when interpreting observational cross-country macro panels, and we highlight five main limitations of our study:

1. *Reverse causality and fiscal-capacity feedback:* A central concern in macro-fiscal studies is that growing tax capacity enables governments to spend more on higher education rather than the other way round (Baumol, 1967; World Bank and UNESCO, 2024; World Bank, 2024). Whilst we adjust for starting tax-to-GDP, starting income and goods tax shares, lagged GDP growth, expected continuous funding $\hat{z}(X_{i,t})$, and explicit pre-treatment fiscal trends (Appendix Table 6), observational conditioning cannot completely rule out multi-decade fiscal feedback loops.

2. *Unobserved concurrent policy reforms:* Our causal estimates rely on conditional ignorability given the 19 baseline characteristics and four follow-up growth controls (Imai et al., 2010; Acharya et al., 2016). If a reformist government simultaneously modernises its electronic tax administration (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022) and increases university funding within the same five-year wave, part of the estimated slope could reflect that concurrent tax reform.

3. *Skilled emigration (brain drain):* In low- and lower-middle-income countries, a substantial share of university graduates emigrate to higher-income economies (Beine et al., 2008). Because emigrants pay direct income taxes in their destination country whilst sending untaxed remittances home (which are spent on domestic goods subject to VAT), high skilled emigration can dampen domestic direct income tax returns in smaller developing nations.

4. *Central government reporting and five-year wave coarseness:* World Bank fiscal series primarily record central government revenue, which can understate state or provincial income taxes in federal countries, and averaging annual data into five-year waves smooths out year-to-year timing.

5. *Continuous dose linearity within tree leaves and winsorisation:* Modelling university funding as a continuous dose $Z_{i,t} \in [0, 3.0]$ avoids the artificial cliff-edge boundary effect of a binary median cutoff (where a country sitting just 0.3 pp below its income-group median, such as New Zealand at 26.1% or Iceland at 26.1% in Appendix Table 7, would be predicted by a binary step-function model to gain a full treatment effect from a token spending nudge across the 26.4% line). Under our continuous MV-BCF specification, closing a 0.31 pp gap (0.031 dose units, costing €0.04 billion per year in New Zealand) yields a proportionally small expected return (€0.004 billion per year), whereas achieving a full +10 pp increase (+1.0 dose unit, costing €1.16 billion per year) is required to realise the full +10 pp slope (€+0.14 billion per year on direct income tax and +0.72 pp on the income tax share). Nevertheless, because $Z_{i,t}$ enters each treatment tree leaf linearly and is winsorised at ±1.5 dose units (±15 pp around the income-group median) to prevent extreme UNESCO ratios from dominating the fit, our model estimates the average marginal slope over the central ±15 pp policy range rather than extrapolating non-linearly to extreme spending shocks.

These limitations point to several promising avenues for future research. First, whilst this paper has focussed exclusively on tertiary education because university cohorts enter the labour market within a 5-to-10-year window, extending continuous-treatment multivariate Bayesian causal forests to 15-to-25-year horizons would allow researchers to compare government spending on early childhood, primary, and secondary schooling (Hanushek and Woessmann, 2012; Hendren and Sprung-Keyser, 2020; Angrist et al., 2021) directly against university funding as younger school cohorts reach adulthood. Second, incorporating bilateral skilled migration data (Beine et al., 2008) directly into the treatment-effect moderators would quantify how brain drain alters the fiscal return to public university subsidies in developing countries. Third, linking university funding cohorts to administrative employer-employee tax registers within individual countries would make it possible to trace graduate earnings and corporate tax payments year by year. To help policy makers and researchers explore our dataset and country-level continuous dose-response estimates interactively, we provide a companion web application in the `app/` directory of the project repository.

Declarations and Statements

Author Contributions (CRediT). **Andrew C. Parnell:** Conceptualisation, Data curation, Formal analysis, Methodology, Software, Visualisation, Writing – original draft, Writing – review and editing.

Data Availability Statement. All raw and processed datasets analysed in this study are derived from publicly available international databases (World Bank WDI, UNESCO UIS, IMF, WGI, SWIID, and OECD) licensed under CC-BY-4.0 and are included in the project repository at `https://github.com/andrewcparnell/education-tax-bcf` (with full SHA-256 checksums and provenance documented in `data/raw/PROVENANCE.md`).

Code Availability Statement. All R and C++ (`RcppArmadillo`) source code, data-cleaning scripts, MCMC samplers, diagnostic scripts, interactive exploration app (`app/`), automated pipeline tests, and figure generation pipelines are openly available under the MIT License at `https://github.com/andrewcparnell/education-tax-bcf` and can be reproduced end-to-end via `bash run_all.sh`.

Ethics, Funding, and Conflict of Interest. This study analyses aggregate sovereign macroeconomic statistics and involves no human subjects or personally identifiable information; institutional ethical approval is not required. This research received no specific external grant funding. As disclosed in Section 2, the author co-developed the Multivariate Bayesian Causal Forest methodology (McJames et al., 2025) evaluated alongside competing estimators in this study; the author declares no financial or other competing interests.

A Mathematical Specification of the Continuous-Treatment MV-BCF Sampler

Let $Y_{i,t+h} = (Y^{(1)}_{i,t+h}, Y^{(2)}_{i,t+h}, Y^{(3)}_{i,t+h}, Y^{(4)}_{i,t+h})^\top \in \mathbb{R}^4$ denote the vector of four disaggregated percentage-point fiscal changes for country i over horizon $h \in \{1, 2\}$ (+5 and +10 years), and let $Z_{i,t} \in [0, 3.0]$ denote the continuous university funding dose. Following Hahn et al. (2020) and McJames et al. (2025), the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) models $Y_{i,t+h}$ as:

$$Y_{i,t+h} = \sum_{m=1}^{M_\mu} g_\mu(X_{i,t}, \hat{z}_{i,t}; T^\mu_m, M^\mu_m) + Z_{i,t} \sum_{l=1}^{M_\tau} g_\tau(W_{i,t}; T^\tau_l, M^\tau_l) + \varepsilon_{i,t+h}, \qquad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma), \quad (3)$$

where $\hat{z}_{i,t} = \hat{\mathbb{E}}[Z_{i,t} \mid X_{i,t}]$ is the random forest expected continuous treatment dose, the prognostic forest uses $M_\mu = 75$ trees with depth-penalty hyperparameters ($\alpha_\mu = 0.95, \beta_\mu = 2.0$), and the treatment-effect forest uses $M_\tau = 30$ trees with more conservative depth-penalty hyperparameters ($\alpha_\tau = 0.25, \beta_\tau = 3.0$) (Hahn et al., 2020). Each prognostic leaf vector $\mu_{m,b} \in \mathbb{R}^4$ and treatment slope leaf vector $\tau_{l,b} \in \mathbb{R}^4$ receives an independent multivariate Gaussian prior $\mathcal{N}_4(0, \Sigma_\mu)$ and $\mathcal{N}_4(0, \Sigma_\tau)$ scaled inversely by the number of trees and the variance of $Z_{i,t}$. Given partial residuals $R^{(l)}_i$ for treatment tree l and precision matrix Σ^{-1}, because $Z_{i,t}$ multiplies $\tau_{l,b}$ linearly inside leaf b, the conjugate treatment leaf posterior is exact multivariate Gaussian with precision $\Omega^{-1}_{\tau,b} = \Sigma^{-1}_\tau + (\sum_{i \in b} Z^2_{i,t})\Sigma^{-1}$ and mean $\Omega_{\tau,b}\Sigma^{-1} \sum_{i \in b} Z_{i,t} R^{(l)}_i$ (with $Z_{i,t} \equiv 1$ for prognostic leaves). During reversible-jump Metropolis-Hastings tree updates (Chipman et al., 2010), a `GROW` move on a terminal node η at depth d_η (with splitting probability $p_{\text{split}}(\eta) = \alpha(1 + d_\eta)^{-\beta}$) is accepted with probability $\min(1, r_{\text{grow}})$, where r_{grow} combines the closed-form Gaussian marginal likelihood ratio with the exact tree prior and transition ratio:

$$r_{\text{grow}} = \frac{p_{\text{prune}}\, b(T)}{p_{\text{grow}}\, w(T^*)} \cdot \frac{p_{\text{split}}(\eta)\big(1 - p_{\text{split}}(\eta_L)\big)\big(1 - p_{\text{split}}(\eta_R)\big)}{1 - p_{\text{split}}(\eta)} \cdot \frac{L(R \mid T^*, \Sigma)}{L(R \mid T, \Sigma)}, \quad (4)$$

where $b(T)$ is the number of terminal nodes in T and $w(T^*)$ is the number of singly internal (`nog`) parent nodes in the proposed tree T^* (with the exact inverse ratio used for `PRUNE` moves). Given full residuals

$E_i = Y_{i,t+h} - \mu_i - \tau_i Z_{i,t}$, the cross-outcome covariance matrix Σ receives a conjugate Inverse-Wishart prior $\mathcal{IW}(\nu_0, \Psi_0)$ with $\nu_0 = 6$ and is updated via $\Sigma \mid E \sim \mathcal{IW}(\nu_0 + N, \Psi_0 + \sum_{i=1}^{N} E_i E_i^\top)$.

B Subgroup Breakdowns, Method Benchmarks, and Robustness Checks

In this appendix we present and interpret the detailed subgroup slopes, competing estimator benchmarks, sensitivity checks, and country-level cost-return calculations referenced in Section 5. Table 4 reports the +10-year subgroup posterior mean slopes and 95% credible intervals per +10 pp funding increase across baseline university enrolment tiers ($N = 244$ observed out of 283), all four World Bank income groups ($N = 283$), and the five OECD financing regimes ($N = 119$ OECD country-waves). Two contrasting lessons emerge from Table 4. First, the compositional shift toward direct income and profit taxation (Column 5) is positive across all enrolment tiers and middle- and high-income groups, rising from +0.64 pp ($P = 0.91$) in elite systems (< 20% enrolment) to +0.75 pp ($P = 0.97$) in massifying systems (20–50% enrolment), +0.84 pp ($P = 0.98$, 95% CI [+0.09, +1.57]) in upper-middle-income economies, and +1.05 to +1.14 pp ($P = 0.93$–0.95) across all five OECD financing regimes regardless of tuition structure. Second, total non-resource tax revenue as a share of GDP (Column 3) follows the opposite pattern across development stages: the largest slopes per +10 pp dose occur in low-income (+0.48 pp, $P = 0.91$) and lower-middle-income (+0.32 pp, $P = 0.93$) economies whose formal tax bases are still expanding, whereas mature high-income economies (+0.19 pp, $P = 0.82$) and the separate OECD-only sub-model (−0.14 to −0.17 pp) already operate near their aggregate tax-to-GDP ceiling and primarily experience a shift in the tax mix toward direct income taxation.

Table 4: Subgroup +10-year posterior mean continuous slopes (in percentage points, pp per +10 pp university spending increase, with 95% CIs on the second line of each cell and posterior probabilities $P = P(\tau > 0 \mid \text{data})$) across starting university enrolment tiers ($N = 244$ observed), World Bank income groups ($N = 283$), and OECD financing regimes ($N = 119$ OECD waves). Subgroups with $N \leq 20$ are small-sample exploratory strata.

Subgroup (+10-Year Horizon, Scaling A)	Waves (Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
1. Elite Enrolment (< 20%)	74 (43)	+0.34 ($P = 0.93$) [−0.11, +0.83]	+0.17 ($P = 0.91$) [−0.09, +0.43]	+0.64 ($P = 0.91$) [−0.21, +1.42]
2. Massifying (20–50%)	73 (42)	+0.23 ($P = 0.89$) [−0.16, +0.69]	+0.12 ($P = 0.88$) [−0.09, +0.32]	+0.75 ($P = 0.97$) [−0.03, +1.47]
3. High-Participation (≥ 50%)	97 (44)	+0.17 ($P = 0.80$) [−0.25, +0.66]	+0.10 ($P = 0.81$) [−0.13, +0.32]	+0.73 ($P = 0.95$) [−0.16, +1.51]
4. Low Income (exploratory)	15 (9)	+0.48 ($P = 0.91$) [−0.24, +1.30]	+0.24 ($P = 0.91$) [−0.12, +0.76]	+0.16 ($P = 0.69$) [−1.96, +1.36]
5. Lower Middle Income	47 (21)	+0.32 ($P = 0.93$) [−0.11, +0.80]	+0.16 ($P = 0.91$) [−0.09, +0.42]	+0.74 ($P = 0.96$) [−0.07, +1.55]
6. Upper Middle Income	76 (35)	+0.23 ($P = 0.89$) [−0.16, +0.68]	+0.13 ($P = 0.90$) [−0.08, +0.33]	+0.84 ($P = 0.98$) [+0.09, +1.57]
7. High Income	145 (47)	+0.19 ($P = 0.82$) [−0.23, +0.66]	+0.10 ($P = 0.81$) [−0.14, +0.31]	+0.68 ($P = 0.94$) [−0.17, +1.44]
8. OECD Nordic High-Public	20 (5)	−0.17 ($P = 0.23$) [−0.62, +0.38]	+0.04 ($P = 0.57$) [−0.34, +0.49]	+1.13 ($P = 0.94$) [−0.18, +2.51]
9. OECD Low-Tuition Public	46 (15)	−0.15 ($P = 0.26$) [−0.58, +0.36]	+0.05 ($P = 0.58$) [−0.33, +0.47]	+1.05 ($P = 0.93$) [−0.26, +2.38]
10. OECD Hybrid Public-Fee	29 (8)	−0.15 ($P = 0.25$) [−0.60, +0.35]	+0.04 ($P = 0.56$) [−0.33, +0.45]	+1.11 ($P = 0.95$) [−0.17, +2.45]
11. OECD High-Tuition High-Sub.	11 (5)	−0.16 ($P = 0.24$) [−0.63, +0.39]	+0.05 ($P = 0.57$) [−0.34, +0.50]	+1.14 ($P = 0.94$) [−0.16, +2.55]
12. OECD High-Tuition Low-Sub.	13 (4)	−0.14 ($P = 0.27$) [−0.59, +0.37]	+0.09 ($P = 0.64$) [−0.31, +0.49]	+1.09 ($P = 0.94$) [−0.30, +2.46]

Table 5 compares our primary joint four-outcome continuous MV-BCF model at the +10-year horizon ($N = 283$, Scaling A) against a specification that omits the follow-up macroeconomic growth controls

(Total Effect M1b), three Bayesian tree ablations (B1 separate univariate BCFs, B2 MV-BCF without the random forest expected treatment $\hat{z}(X)$, and B3 single-forest BART), and three frequentist panel estimators (B4 linear local projections, Jordà 2005; B5 country-and-wave two-way fixed effects; and B6 5-fold country-blocked cross-fitted double machine learning, Chernozhukov et al. 2018). Three methodological lessons stand out from Table 5. First, comparing the first two rows shows that omitting the follow-up macroeconomic growth controls strengthens the 10-year slopes on total non-resource tax/GDP (+0.30 pp, $P = 0.95$ vs +0.24 pp, $P = 0.89$) and direct income tax/GDP (+0.16 pp, $P = 0.96$ vs +0.12 pp, $P = 0.90$) whilst preserving a positive shift in the direct income tax share (+0.54 pp, $P = 0.92$ vs +0.71 pp, $P = 0.96$), confirming that part of the total fiscal return operates through broader economic growth whilst the compositional shift toward direct income taxation is a direct structural effect. Second, amongst the Bayesian tree specifications (B1–B3), univariate BCFs (B1) and MV-BCF without $\hat{z}(X)$ (B2) yield positive slopes closely aligned with our primary model (with the joint covariance matrix Σ in our primary model improving posterior precision on the direct income tax share relative to independent univariate BCFs: SD = 0.385 vs 0.462 pp), whereas single-forest BART (B3) shrinks the continuous treatment slopes toward zero (+0.03 pp on direct income tax/GDP and +0.34 pp on the income tax share), illustrating the regularisation-induced confounding described by Hahn et al. (2020) when prognostic and treatment trees are not given separate priors. Third, comparing across the frequentist panel estimators (B4–B6) shows that whilst headline total tax-to-GDP is sensitive to two-way fixed effects (B5, −0.21 pp vs +0.59 pp in linear local projections B4 and +0.41 pp in double machine learning B6), both direct income tax/GDP (+0.06 to +0.18 pp) and the direct income tax share (+0.07 to +0.51 pp, and +0.62 pp, $P = 0.98$ when country fixed effects are incorporated via within-country demeaning inside MV-BCF in R3b below) remain positive across every method (note that linear double machine learning B6 weights observations by residual treatment variance $\tilde{Z}_{i,t}^2$, which places greater weight on volatile lower-income country-waves where the compositional tax-share slope is smallest at +0.16 pp, whereas MV-BCF estimates the unweighted population-average slope $\bar{\tau} = N^{-1} \sum_{i=1}^{N} \tau(W_{i,t})$).

Table 5: Method benchmark and ablation comparison for the continuous +10 pp university spending dose (Scaling A) at the +10-year horizon on the identical common sample ($N = 283$, 112 countries). Posterior means, 95% Bayesian CIs, and posterior probabilities P are shown for the Bayesian tree specifications; frequentist point estimates, 95% confidence intervals, and one-sided p-value complements $P_{\text{freq}} = 1 - p_{\text{one-sided}}$ are shown for B4–B6.

Estimator / Specification (+10y, Scaling A)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)	Goods Tax Share (pp)
Primary: Joint MV-BCF (Full Controls, M2b)	+0.24 ($P = 0.89$) [−0.15, +0.70]	+0.12 ($P = 0.90$) [−0.08, +0.33]	+0.71 ($P = 0.96$) [−0.08, +1.42]	−0.42 ($P = 0.14$) [−1.19, +0.34]
MV-BCF Without Growth Controls (M1b)	+0.30 ($P = 0.95$) [−0.05, +0.65]	+0.16 ($P = 0.96$) [−0.02, +0.34]	+0.54 ($P = 0.92$) [−0.19, +1.33]	−0.65 ($P = 0.04$) [−1.45, +0.10]
B1: Univariate BCFs ($D = 1$)	+0.45 ($P = 0.97$) [−0.004, +0.86]	+0.17 ($P = 0.94$) [−0.03, +0.39]	+0.71 ($P = 0.93$) [−0.21, +1.63]	−0.47 ($P = 0.15$) [−1.38, +0.44]
B2: MV-BCF Without $\hat{z}(X)$	+0.34 ($P = 0.95$) [−0.05, +0.73]	+0.14 ($P = 0.90$) [−0.08, +0.35]	+0.59 ($P = 0.94$) [−0.11, +1.35]	−0.51 ($P = 0.10$) [−1.23, +0.34]
B3: Single-Forest BART	+0.15 ($P = 0.67$) [−0.53, +0.85]	+0.03 ($P = 0.57$) [−0.30, +0.38]	+0.34 ($P = 0.68$) [−0.93, +1.62]	−0.09 ($P = 0.43$) [−1.39, +1.08]
B4: Linear Local Projections	+0.59 ($P_{\text{freq}} = 0.99$) [+0.07, +1.12]	+0.18 ($P_{\text{freq}} = 0.90$) [−0.09, +0.44]	+0.51 ($P_{\text{freq}} = 0.82$) [−0.58, +1.59]	+0.45 ($P_{\text{freq}} = 0.78$) [−0.70, +1.59]
B5: Two-Way Fixed Effects	−0.21 ($P_{\text{freq}} = 0.30$) [−1.01, +0.59]	+0.08 ($P_{\text{freq}} = 0.61$) [−0.50, +0.67]	+0.50 ($P_{\text{freq}} = 0.70$) [−1.40, +2.41]	−0.32 ($P_{\text{freq}} = 0.37$) [−2.27, +1.63]
B6: Cross-Fitted Double ML	+0.41 ($P_{\text{freq}} = 0.95$) [−0.09, +0.91]	+0.06 ($P_{\text{freq}} = 0.69$) [−0.18, +0.29]	+0.07 ($P_{\text{freq}} = 0.56$) [−0.94, +1.08]	+0.56 ($P_{\text{freq}} = 0.80$) [−0.73, +1.84]

Table 6 reports 15 sensitivity and falsification checks (R1 to R14) designed to test every data-construction and modelling decision in our continuous-treatment pipeline. First, the +10-year positive slopes on non-resource tax/GDP, direct income tax/GDP, and the direct income tax share remain consistent when we use raw unadjusted WDI tax-to-GDP without natural resource rent deductions (R1, and +0.24 pp on total tax/GDP, $P = 0.89$, and +0.70 pp on the income tax share, $P = 0.96$, when the 150% pre-winsorisation cap is omitted in R1b of `outputs/tables/table4_robustness_checks.csv`), collapse the panel into non-overlapping 10-year decadal waves (R2, +0.82 pp on the income tax share, $P = 0.93$),

sweep out time-invariant country fixed effects via within-country demeaning (R3a at +5 years, +0.44 pp on the income tax share, $P = 0.97$; and R3b at +10 years, +0.12 pp on direct income tax/GDP, $P = 0.94$, and +0.62 pp on the income tax share, 95% CI [+0.01, +1.31], $P = 0.98$), drop all observations with any missing baseline covariate (R4, +0.26 pp on total tax/GDP, $P = 0.91$, and +0.66 pp on the income tax share, $P = 0.96$), use a more diffuse treatment-effect tree prior (R6, +0.30 pp on total tax/GDP, $P = 0.96$, and +0.16 pp on direct income tax/GDP, $P = 0.97$), perturb SWIID Gini inequality with its reported standard errors (R7), use row-wise out-of-bag random forest predictions for $\hat{z}(X)$ (R8), apply two-stage sequential g-estimation with a linear additive first-stage demediation regression (R9; Acharya et al. 2016, +0.46 pp on total tax/GDP, $P = 0.99$, +0.20 pp on direct income tax/GDP, $P = 0.97$, and +0.85 pp on the income tax share, $P = 0.98$), control explicitly for lagged pre-treatment tax trends $\Delta Y_{t-1 \to t}$ (R10, +0.28 pp on total tax/GDP, $P = 0.96$, and +0.16 pp on direct income tax/GDP, $P = 0.94$), or replace the goods tax share with goods tax as a percentage of GDP (R11). Second, our pre-treatment falsification test (R5), which checks whether continuous university funding at wave t "predicts" prior tax changes over $t - 1 \to t$, yields near-zero and statistically indistinct slopes (+0.04 pp on direct income tax/GDP, $P = 0.67$, and +0.001 pp on the direct income tax share, $P = 0.51$), reassuring us that treatment assignment is not simply tracking pre-existing fiscal booms. Third, restricting the sample to pre-COVID waves before 2020 (R12, $N = 187$) preserves positive point estimates (+0.23 pp on total tax/GDP, $P = 0.82$, and +0.46 pp on the direct income tax share, $P = 0.82$), fitting strictly on the 80% training countries (R13, Folds 1 to 4, $N = 229$ across 90 countries) yields +0.42 pp on total tax/GDP ($P = 0.99$), +0.22 pp on direct income tax/GDP ($P = 0.98$), and +0.64 pp on the income tax share ($P = 0.92$), and the legacy binary median-cutoff specification (R14, $Z_{i,t} \in \{0, 1\}$, where $\mathrm{SD}(Z) = 0.50$ so the binary jump corresponds to roughly two +10 pp continuous dose units) yields +0.58 pp on total tax/GDP ($P = 0.93$), +0.27 pp on direct income tax/GDP ($P = 0.92$), and +1.18 pp on the direct income tax share ($P = 0.95$).

Finally, Table 7 lists all 44 countries from our 10-year modelling sample that sit below their income-group training median ($Z_{i,t} < 1.5$) in the most recent treatment wave (2015–2019), ordered within each World Bank income group from closest to the median to furthest below it (full data in `outputs/tables/tableA4_latest_z0_countries_costs_and_returns.csv`). Table 7 illustrates why our continuous-treatment specification is much more realistic for country-level policy evaluation than a binary threshold model. Six countries (marked with *: New Zealand, Iceland, Uruguay, Romania, Italy, and Slovenia) sit within 2.5 pp of their income-group median (26.4% for high income); because our continuous MV-BCF model scales the expected fiscal return smoothly with the size of the funding increase, closing a tiny gap to the median (such as +0.31 pp in New Zealand costing €0.04 billion per year or +0.35 pp in Iceland costing €0.005 billion per year) yields a proportionally tiny expected return (first number in Column 5: €+0.00 billion per year when rounded to two decimal places), whereas a full +10 pp (+1.0 dose unit) expansion costs €1.16 billion per year in New Zealand and €0.13 billion per year in Iceland and yields +0.06 to +0.09 pp of GDP in direct income tax (second number in Column 5: €+0.14 billion and €+0.02 billion per year) alongside a +0.72 to +0.77 pp shift in the direct income tax share (Column 6). For countries sitting roughly one full +10 pp dose unit below their income-group median (such as Ireland at 17.3% with a 9.1 pp gap costing €2.16 billion per year to reach the median or €2.37 billion per year for a +10 pp boost, Korea at 14.9%, Argentina at 16.7%, and Belarus at 17.0%), closing the gap to the median and adding a +10 pp dose represent comparable investments whose expected 10-year direct income tax returns are reported with full 95% credible intervals in Columns 4 to 6 (for deep-gap developing countries sitting more than 15 pp below their group median, the gap return in Column 5 is evaluated at the model's +15 pp, or 1.5 dose unit, winsorisation boundary rather than extrapolating linearly beyond the support of $Z_{i,t} \in [0, 3.0]$).

C Continuous Treatment Orthogonalisation, Residual Covariance, and MCMC Diagnostics

Table 8 summarises how our random forest expected continuous treatment $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ orthogonalises the continuous university funding dose against all 19 pre-treatment covariates (full table in

Table 6: Robustness, identification, and falsification checks across the primary tax outcomes (with 95% CIs on the second line of each cell and posterior probabilities $P = P(\tau > 0 \mid \text{data})$). Checks `R1`–`R13` report continuous slopes per +10 pp funding dose (Scaling A); `R14` reports the legacy binary median-cutoff contrast ($Z \in \{0, 1\}$).

Sensitivity Check	Horizon (N, Countries)	Total Tax / GDP (pp)	Direct Income Tax / GDP (pp)	Direct Income Tax Share (pp)
`R1`. **Unadjusted WDI Tax / GDP**	+10y (283, 112)	+0.25 ($P = 0.89$) [−0.16, +0.63]	+0.15 ($P = 0.92$) [−0.07, +0.38]	+0.45 ($P = 0.84$) [−0.34, +1.28]
`R2`. **Non-Overlapping 10y Waves**	+10y (154, 105)	+0.32 ($P = 0.89$) [−0.17, +0.85]	+0.12 ($P = 0.79$) [−0.16, +0.40]	+0.82 ($P = 0.93$) [−0.30, +1.83]
`R3a`. **Country Fixed Effects (+5y)**	+5y (371, 121)	+0.01 ($P = 0.53$) [−0.21, +0.23]	+0.06 ($P = 0.86$) [−0.05, +0.17]	+0.44 ($P = 0.97$) [−0.03, +0.90]
`R3b`. **Country Fixed Effects (+10y)**	+10y (283, 112)	+0.18 ($P = 0.86$) [−0.17, +0.50]	+0.12 ($P = 0.94$) [−0.04, +0.26]	+0.62 ($P = 0.98$) [+0.01, +1.31]
`R4`. **Strictly Observed Covariates**	+10y (238, 107)	+0.26 ($P = 0.91$) [−0.11, +0.63]	+0.11 ($P = 0.84$) [−0.09, +0.35]	+0.66 ($P = 0.96$) [−0.06, +1.47]
`R5`. **Placebo ($\Delta Y_{t-1\to t}$)**	Pre-t (347, 117)	+0.09 ($P = 0.74$) [−0.19, +0.36]	+0.04 ($P = 0.67$) [−0.11, +0.18]	+0.00 ($P = 0.51$) [−0.59, +0.56]
`R6`. **Diffuse τ Prior**	+10y (283, 112)	+0.30 ($P = 0.96$) [−0.02, +0.67]	+0.16 ($P = 0.97$) [−0.01, +0.34]	+0.61 ($P = 0.94$) [−0.17, +1.42]
`R7`. **SWIID Error Perturbation**	+10y (283, 112)	+0.33 ($P = 0.95$) [−0.06, +0.68]	+0.14 ($P = 0.89$) [−0.09, +0.35]	+0.55 ($P = 0.93$) [−0.20, +1.29]
`R8`. **Row-Wise OOB RF $\hat{z}(X)$**	+10y (283, 112)	+0.31 ($P = 0.93$) [−0.10, +0.69]	+0.13 ($P = 0.90$) [−0.08, +0.33]	+0.62 ($P = 0.94$) [−0.19, +1.40]
`R9`. **Sequential G-Estimation**	+10y (283, 112)	+0.46 ($P = 0.99$) [+0.09, +0.85]	+0.20 ($P = 0.97$) [−0.01, +0.41]	+0.85 ($P = 0.98$) [+0.05, +1.66]
`R10`. **Pre-Trend Controls ($\Delta Y_{t-1\to t}$)**	+10y (283, 112)	+0.28 ($P = 0.96$) [−0.03, +0.61]	+0.16 ($P = 0.94$) [−0.03, +0.36]	+0.58 ($P = 0.93$) [−0.18, +1.42]
`R11`. **Goods Tax / GDP Check**	+10y (283, 112)	+0.30 ($P = 0.95$) [−0.04, +0.63]	+0.15 ($P = 0.92$) [−0.06, +0.35]	+0.50 ($P = 0.90$) [−0.30, +1.37]
`R12`. **Pre-COVID Sample (< 2020)**	+10y (187, 90)	+0.23 ($P = 0.82$) [−0.27, +0.70]	+0.08 ($P = 0.70$) [−0.20, +0.36]	+0.46 ($P = 0.82$) [−0.49, +1.44]
`R13`. **80% Training Set Only**	+10y (229, 90)	+0.42 ($P = 0.99$) [+0.06, +0.79]	+0.22 ($P = 0.98$) [+0.001, +0.44]	+0.64 ($P = 0.92$) [−0.31, +1.63]
`R14`. **Legacy Binary Cutoff ($Z \in \{0, 1\}$)**	+10y (283, 112)	+0.58 ($P = 0.93$) [−0.22, +1.37]	+0.27 ($P = 0.92$) [−0.10, +0.68]	+1.18 ($P = 0.95$) [−0.25, +2.62]

Table 7: Countries sitting below their income-group training median ($Z_{i,t} < 1.5$) in the most recent treatment wave (2015–2019, $N = 44$ countries with +10-year continuous MV-BCF estimates), showing their current per-student spending ratio (% of GDP per person) and gap to the Scaling A income-group median (26.4% High, 27.6% Upper Mid., 68.9% Lower Mid., 150.0% Low), annual public cost to close the gap in % of GDP and constant 2021 euros (followed after the slash by the annual cost of a full +10 pp dose), and estimated +10-year posterior mean returns (with 95% CIs) per +10 pp dose on direct income tax/GDP (pp), annual direct income tax revenue in € billions/yr (showing the proportional return to closing the median gap first, capped at +15 pp or 1.5 dose units for deep-gap countries, followed in brackets by the return and 95% CI for a full +10 pp dose), and the direct income tax share (pp). Near-boundary countries within 2.5 pp of the median are marked *.

Country (Income Group)	Current Ratio (Gap to Median)	Annual Cost: Gap / +10 pp % GDP (€B Gap / €B +10pp)	+10y Inc. Tax / GDP per +10pp [95% CI]	+10y Annual Inc. Tax Return Gap (€B +10pp) [95% CI]	+10y Inc. Tax Share per +10pp [95% CI]
New Zealand* (High)	26.1% (+0.31 pp)	0.02% (€0.04B / €1.16B)	+0.06 [−0.21, +0.32]	€+0.00 (€+0.14) [−0.49, +0.73]	+0.72 [−0.20, +1.55]
Iceland* (High)	26.1% (+0.35 pp)	0.02% (€0.005B / €0.13B)	+0.09 [−0.16, +0.34]	€+0.00 (€+0.02) [−0.04, +0.08]	+0.77 [−0.09, +1.60]
Uruguay* (High)	25.3% (+1.1 pp)	0.05% (€0.05B / €0.41B)	+0.17 [−0.10, +0.48]	€+0.02 (€+0.16) [−0.09, +0.45]	+0.85 [−0.04, +1.82]
Romania* (High)	25.1% (+1.3 pp)	0.04% (€0.26B / €1.96B)	+0.06 [−0.19, +0.29]	€+0.06 (€+0.42) [−1.31, +1.95]	+0.72 [−0.14, +1.51]
Italy* (High)	24.8% (+1.6 pp)	0.05% (€1.35B / €8.47B)	+0.06 [−0.20, +0.29]	€+0.27 (€+1.68) [−5.56, +8.10]	+0.72 [−0.16, +1.51]
Slovenia* (High)	24.1% (+2.3 pp)	0.09% (€0.09B / €0.37B)	+0.06 [−0.19, +0.30]	€+0.01 (€+0.06) [−0.17, +0.27]	+0.73 [−0.15, +1.52]
Singapore (High)	23.5% (+2.9 pp)	0.13% (€0.92B / €3.23B)	+0.61 [−0.06, +1.27]	€+1.21 (€+4.22) [−0.39, +8.74]	−1.07 [−4.16, +2.09]
Hungary (High)	23.3% (+3.1 pp)	0.09% (€0.31B / €1.02B)	+0.06 [−0.19, +0.29]	€+0.07 (€+0.22) [−0.67, +1.00]	+0.72 [−0.14, +1.50]
Latvia (High)	22.6% (+3.9 pp)	0.16% (€0.10B / €0.26B)	+0.06 [−0.19, +0.29]	€+0.01 (€+0.04) [−0.12, +0.18]	+0.71 [−0.18, +1.51]
Spain (High)	22.4% (+4.0 pp)	0.17% (€3.37B / €8.34B)	+0.06 [−0.19, +0.29]	€+0.52 (€+1.30) [−3.81, +5.90]	+0.73 [−0.13, +1.51]
Macao SAR, China (High)	22.3% (+4.1 pp)	0.21% (€0.11B / €0.27B)	+0.38 [−0.28, +1.05]	€+0.08 (€+0.20) [−0.15, +0.54]	−0.19 [−3.00, +2.61]
Lithuania (High)	21.4% (+5.0 pp)	0.22% (€0.27B / €0.54B)	+0.06 [−0.19, +0.29]	€+0.04 (€+0.07) [−0.23, +0.35]	+0.72 [−0.14, +1.51]
Czechia (High)	20.7% (+5.7 pp)	0.17% (€0.80B / €1.39B)	+0.06 [−0.19, +0.29]	€+0.17 (€+0.30) [−0.89, +1.38]	+0.73 [−0.14, +1.52]
United States (High)	20.5% (+5.9 pp)	0.42% (€92.72B / €156.47B)	+0.06 [−0.23, +0.31]	€+7.88 (€+13.28) [−50.56, +68.14]	+0.71 [−0.20, +1.52]
Australia (High)	19.6% (+6.8 pp)	0.54% (€7.79B / €11.54B)	+0.06 [−0.22, +0.30]	€+0.58 (€+0.85) [−3.12, +4.22]	+0.72 [−0.19, +1.51]
Chile (High)	19.6% (+6.9 pp)	0.48% (€2.49B / €3.63B)	+0.09 [−0.17, +0.34]	€+0.32 (€+0.47) [−0.89, +1.77]	+0.76 [−0.10, +1.59]
Russian Federation (High)	19.3% (+7.1 pp)	0.33% (€16.83B / €23.72B)	+0.16 [−0.35, +0.68]	€+5.89 (€+8.30) [−18.18, +35.41]	+0.97 [−1.31, +3.10]
Israel (High)	18.7% (+7.7 pp)	0.38% (€1.57B / €2.05B)	+0.17 [−0.10, +0.48]	€+0.55 (€+0.71) [−0.41, +1.98]	+0.87 [+0.00, +1.85]
Ireland (High)	17.3% (+9.1 pp)	0.39% (€2.16B / €2.37B)	+0.06 [−0.19, +0.29]	€+0.33 (€+0.36) [−1.03, +1.62]	+0.73 [−0.12, +1.51]
Korea, Rep. (High)	14.9% (+11.5 pp)	0.67% (€16.73B / €14.52B)	+0.07 [−0.19, +0.30]	€+1.92 (€+1.67) [−4.89, +7.51]	+0.74 [−0.14, +1.59]
Greece (High)	11.2% (+15.2 pp)	1.20% (€4.03B / €2.65B)	+0.06 [−0.19, +0.29]	€+0.32 (€+0.21) [−0.64, +0.97]	+0.72 [−0.14, +1.51]
Malaysia (Upper Mid.)	24.8% (+2.8 pp)	0.11% (€1.14B / €4.09B)	+0.09 [−0.19, +0.35]	€+0.26 (€+0.94) [−1.94, +3.58]	+0.75 [−0.13, +1.59]
Jordan (Upper Mid.)	23.7% (+3.9 pp)	0.11% (€0.12B / €0.30B)	+0.16 [−0.10, +0.45]	€+0.07 (€+0.17) [−0.10, +0.47]	+0.81 [−0.08, +1.71]
Indonesia (Upper Mid.)	20.8% (+6.8 pp)	0.08% (€2.78B / €4.08B)	+0.16 [−0.14, +0.45]	€+3.78 (€+5.53) [−4.74, +15.59]	+0.82 [−0.07, +1.71]
Colombia (Upper Mid.)	20.6% (+7.0 pp)	0.34% (€2.89B / €4.13B)	+0.12 [−0.15, +0.39]	€+0.69 (€+0.99) [−1.24, +3.29]	+0.80 [−0.06, +1.67]
Azerbaijan (Upper Mid.)	20.2% (+7.4 pp)	0.16% (€0.31B / €0.41B)	+0.05 [−0.25, +0.32]	€+0.08 (€+0.11) [−0.49, +0.61]	+0.73 [−0.25, +1.86]
Guatemala (Upper Mid.)	18.1% (+9.6 pp)	0.22% (€0.44B / €0.46B)	+0.17 [−0.10, +0.46]	€+0.31 (€+0.33) [−0.19, +0.91]	+0.84 [−0.04, +1.75]
Belarus (Upper Mid.)	17.0% (+10.6 pp)	0.50% (€1.18B / €1.11B)	+0.19 [−0.22, +0.62]	€+0.48 (€+0.46) [−0.51, +1.45]	+1.29 [−0.47, +2.98]
Argentina (Upper Mid.)	16.7% (+10.9 pp)	0.72% (€7.93B / €7.32B)	+0.17 [−0.10, +0.48]	€+2.07 (€+1.90) [−1.08, +5.30]	+0.86 [−0.01, +1.84]
Albania (Upper Mid.)	13.8% (+13.8 pp)	0.71% (€0.31B / €0.22B)	+0.06 [−0.20, +0.30]	€+0.04 (€+0.03) [−0.09, +0.13]	+0.71 [−0.22, +1.52]
El Salvador (Upper Mid.)	11.2% (+16.4 pp)	0.47% (€0.29B / €0.18B)	+0.17 [−0.10, +0.46]	€+0.16 (€+0.11) [−0.06, +0.29]	+0.85 [+0.00, +1.79]
Georgia (Upper Mid.)	10.7% (+16.9 pp)	0.58% (€0.41B / €0.25B)	+0.06 [−0.19, +0.29]	€+0.07 (€+0.04) [−0.14, +0.21]	+0.73 [−0.12, +1.51]
Peru (Upper Mid.)	10.6% (+17.0 pp)	1.01% (€4.69B / €2.76B)	+0.17 [−0.10, +0.48]	€+1.20 (€+0.80) [−0.46, +2.22]	+0.86 [+0.01, +1.84]
Mauritius (Upper Mid.)	10.4% (+17.2 pp)	0.50% (€0.14B / €0.08B)	+0.05 [−0.27, +0.31]	€+0.02 (€+0.01) [−0.08, +0.09]	+0.73 [−0.29, +1.79]
Armenia (Upper Mid.)	9.8% (+17.8 pp)	0.59% (€0.29B / €0.16B)	+0.06 [−0.19, +0.29]	€+0.04 (€+0.03) [−0.09, +0.14]	+0.72 [−0.15, +1.51]
Kazakhstan (Upper Mid.)	9.3% (+18.3 pp)	0.64% (€4.02B / €2.20B)	+0.06 [−0.19, +0.30]	€+0.61 (€+0.40) [−1.20, +1.86]	+0.73 [−0.13, +1.51]
Mongolia (Upper Mid.)	7.9% (+19.7 pp)	1.07% (€0.53B / €0.27B)	+0.13 [−0.13, +0.41]	€+0.10 (€+0.07) [−0.06, +0.20]	+0.80 [−0.06, +1.69]
Tunisia (Lower Mid.)	54.7% (+14.2 pp)	0.38% (€0.54B / €0.38B)	+0.11 [−0.15, +0.37]	€+0.22 (€+0.16) [−0.21, +0.52]	+0.76 [−0.09, +1.59]
Lesotho (Lower Mid.)	44.7% (+24.2 pp)	0.31% (€0.02B / €0.01B)	+0.25 [−0.13, +0.66]	€+0.02 (€+0.01) [−0.01, +0.04]	+0.03 [−1.60, +1.48]
Honduras (Lower Mid.)	40.9% (+28.0 pp)	0.57% (€0.34B / €0.12B)	+0.17 [−0.10, +0.46]	€+0.15 (€+0.10) [−0.06, +0.28]	+0.84 [−0.04, +1.76]
Bangladesh (Lower Mid.)	29.9% (+39.0 pp)	0.74% (€8.99B / €2.31B)	+0.17 [−0.10, +0.46]	€+3.03 (€+2.02) [−1.20, +5.56]	+0.83 [−0.04, +1.75]
Nepal (Lower Mid.)	24.2% (+44.7 pp)	0.81% (€1.04B / €0.23B)	+0.16 [−0.10, +0.45]	€+0.32 (€+0.21) [−0.13, +0.58]	+0.82 [−0.06, +1.73]
Kyrgyz Republic (Lower Mid.)	4.9% (+64.0 pp)	2.33% (€0.93B / €0.15B)	+0.17 [−0.10, +0.47]	€+0.10 (€+0.07) [−0.04, +0.19]	+0.85 [−0.07, +1.84]
Rwanda (Low)	102.8% (+47.2 pp)	0.31% (€0.12B / €0.02B)	+0.21 [−0.15, +0.76]	€+0.12 (€+0.08) [−0.06, +0.28]	+0.59 [−1.67, +1.70]

outputs/tables/tableA3_propensity_overlap_and_balance.csv). Before conditioning on $\hat{z}(X_{i,t})$, country-waves with above-median university funding doses ($Z_{i,t} \geq 1.5$) have higher total education spending as a share of GDP (5.26% vs 3.84%, raw correlation $r = +0.540$), higher baseline non-resource tax-to-GDP (18.57% vs 15.65%, $r = +0.325$), higher baseline direct income tax share (36.57% vs 32.93%, $r = +0.196$), and slower lagged GDP per person growth (1.53% vs 3.49%, $r = -0.297$) than country-waves with lower funding doses ($Z_{i,t} < 1.5$). After residualising $Z_{i,t}$ on the country-blocked random forest prediction $\hat{z}(X_{i,t})$, the mean absolute correlation between the residual treatment $Z_{i,t} - \hat{z}(X_{i,t})$ and all 19 baseline covariates drops by more than 60% from 0.157 (maximum 0.540) to 0.061 (maximum 0.213 for Scaling A, and from 0.140 to 0.056 for Scaling B). This confirms that the random forest first stage captures the systematic pre-treatment predictors of continuous funding intensity and that including $\hat{z}(X_{i,t})$ inside the prognostic forest $\mu(\cdot)$ effectively orthogonalises the treatment dose against baseline confounders.

Table 8: Continuous treatment covariate correlations before and after residualising on the random forest expected treatment dose $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ across key baseline predictors and the 19-covariate summary (+10-year sample, $N = 283$).

Baseline Pre-Treatment Covariate	Low Dose ($Z < 1.5$)	High Dose ($Z \geq 1.5$)	Raw Corr. (Scaling A)	Resid. Corr. (Scaling A)	Raw Corr. (Scaling B)	Resid. Corr. (Scaling B)
Total Education Spend (% GDP)	3.84	5.26	+0.540	+0.166	+0.516	+0.179
Baseline Non-Resource Tax / GDP (%)	15.65	18.57	+0.325	+0.044	+0.275	+0.037
Lagged GDP per Person Growth (%)	3.49	1.53	−0.297	−0.134	−0.317	−0.138
Baseline Direct Income Tax Share (%)	32.93	36.57	+0.196	+0.028	+0.176	+0.015
Log Real GDP per Person (PPP)	9.75	9.74	+0.142	+0.025	+0.074	−0.015
Gross Tertiary Enrolment (%)	45.33	36.23	−0.090	−0.131	−0.119	−0.127
All 19 Covariates (Mean / Max $\lvert r \rvert$)	–	–	**0.157 / 0.540**	**0.061 / 0.213**	**0.140 / 0.516**	**0.056 / 0.202**

Table 9 reports the posterior mean residual standard deviations and all six cross-outcome residual correlations from the continuous-treatment MV-BCF model. These posterior correlations show why modelling all four tax outcomes jointly in a Multivariate BCF is preferable to fitting four independent regressions. Unobserved fiscal shocks to total non-resource tax/GDP and direct income tax/GDP are strongly positively correlated ($r = +0.62$ at +5 years and +0.65 at +10 years), as are shocks to direct income tax/GDP and the direct income tax share ($r = +0.71$ and +0.75), whereas shocks to the direct income tax share and the goods and services tax share are strongly negatively correlated ($r = -0.46$ at +5 years and −0.59 at +10 years) because the two revenue shares compete within the same national budget. Estimating the full 4×4 covariance matrix Σ via the Inverse-Wishart Gibbs step allows the continuous tree updates to borrow strength across these strongly correlated fiscal innovations.

Table 9: Posterior mean cross-outcome residual correlations (with 95% CIs) and residual standard deviations from the four-outcome continuous-treatment Multivariate Bayesian Causal Forest (Scaling A).

Residual Parameter Pair (MV-BCF, Scaling A)	+5y Mean [95% CI]	+10y Mean [95% CI]
1. Non-Res. Tax / GDP ↔ Income Tax / GDP	+0.62 [+0.54, +0.69]	+0.65 [+0.56, +0.73]
2. Income Tax / GDP ↔ Income Tax Share	+0.71 [+0.65, +0.77]	+0.75 [+0.67, +0.81]
3. Income Tax Share ↔ Goods Tax Share	−0.46 [−0.56, −0.36]	−0.59 [−0.69, −0.48]
4. Income Tax / GDP ↔ Goods Tax Share	−0.24 [−0.35, −0.12]	−0.25 [−0.40, −0.11]
5. Non-Res. Tax / GDP ↔ Goods Tax Share	+0.20 [+0.07, +0.32]	+0.33 [+0.19, +0.46]
6. Non-Res. Tax / GDP ↔ Income Tax Share	−0.03 [−0.14, +0.09]	+0.08 [−0.07, +0.23]
7. Residual SDs ($\sigma_1, \sigma_2, \sigma_3, \sigma_4$)	1.80, 1.00, 4.08, 4.21 pp	2.51, 1.39, 5.14, 4.80 pp

Table 10 and Figure 8 present the post-burn-in MCMC convergence diagnostics and posterior predictive check (PPC) fit metrics across the +5-year and +10-year continuous-treatment MV-BCF models (1,250 retained draws). Across all eight horizon-by-outcome parameters in Table 10, split-chain $\hat{R}$ values lie between 0.999 and 1.030 (well below the standard 1.05 convergence threshold), effective sample sizes (ESS) range from 65 to 120, and Monte Carlo standard errors (MCSE) are small (0.006 to 0.046 pp, less than

13% of the corresponding posterior standard deviations). Furthermore, in-sample posterior predictive check R^2 values range from 0.247 to 0.641 across the macro-panel difference outcomes, empirical 95% posterior predictive interval coverage (`Cov95`) lies between 95.4% and 97.9%, and 93.8% to 95.8% of standardised residuals fall within ±2 standard deviations. Together with the stationary, well-mixed trace plots in Figure 8, these diagnostics confirm that our continuous-treatment C++ Gibbs sampler converges reliably and is well calibrated to the observed distribution of fiscal changes.

Table 10: Post-burn-in MCMC convergence diagnostics (split-chain $\hat{R}$, effective sample size ESS, Monte Carlo standard error MCSE, and Geweke z-score) and in-sample posterior predictive check (PPC) R^2, 95% CI coverage, and share of standardised residuals within ±2 SD across the +5-year and +10-year continuous MV-BCF models (Scaling A).

Model & Outcome (Scaling A)	Horizon (N)	Mean (SD)	$\hat{R}$	ESS	MCSE	Geweke	PPC R^2	Cov95	$\lvert e^* \rvert \leq 2$SD
1. MV-BCF Tax / GDP	+5y (371)	−0.055 (0.113)	**1.026**	91	0.012	−1.67	0.396	96.2%	93.8%
2. MV-BCF Inc. Tax / GDP	+5y (371)	+0.053 (0.061)	**1.030**	92	0.006	+1.80	0.247	96.2%	95.4%
3. MV-BCF Inc. Tax Share	+5y (371)	+0.365 (0.244)	**1.023**	120	0.022	+0.64	0.332	95.4%	94.1%
4. MV-BCF Goods Share	+5y (371)	+0.098 (0.293)	**1.011**	75	0.034	+0.79	0.426	97.0%	94.6%
5. MV-BCF Tax / GDP	+10y (283)	+0.236 (0.200)	**1.003**	65	0.025	−0.29	0.423	95.8%	94.0%
6. MV-BCF Inc. Tax / GDP	+10y (283)	+0.124 (0.103)	**1.002**	109	0.010	−0.34	0.299	96.5%	95.4%
7. MV-BCF Inc. Tax Share	+10y (283)	+0.707 (0.385)	**1.024**	100	0.038	−1.24	0.460	96.8%	94.7%
8. MV-BCF Goods Share	+10y (283)	−0.422 (0.390)	**0.999**	72	0.046	−1.75	0.641	97.9%	95.8%

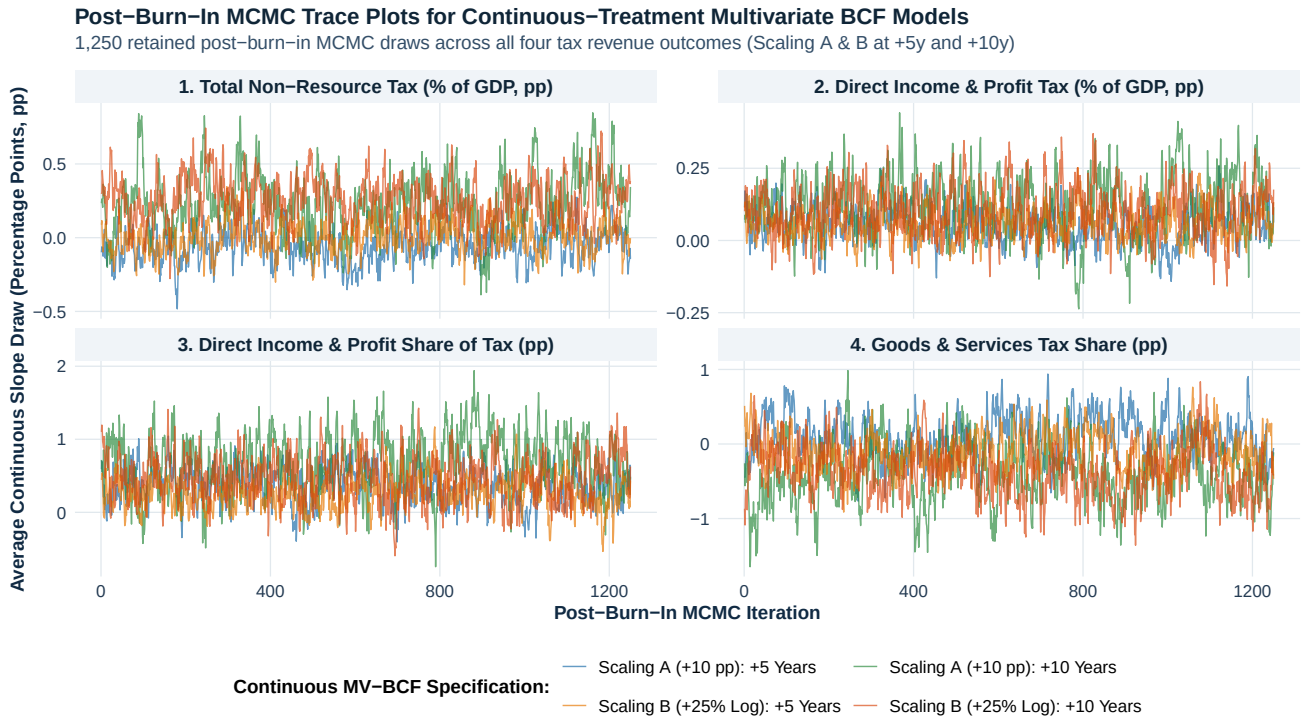


Figure 8: Post-burn-in MCMC trace plots across all four tax outcomes from the +5y and +10y continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF, Scaling A) sampler.

D Comparison of Linear Ratio Scaling A and Relative Logarithmic Scaling B

As discussed in Section 3.3, a continuous per-student university spending dose can be constructed either on a linear percentage-point scale within each World Bank income group (Scaling A: $\Delta Z = +1.0$ per +10 pp of GDP per person per student around the training-set group median $m_{g(i,t)}$, winsorised at ±1.5 units and shifted by +1.5 so $Z_{i,t} \in [0, 3.0]$ with $\mathrm{SD}(Z) = 1.05$) or on a relative logarithmic scale (Scaling

B: $Z_{i,t}^{(\mathrm{B})} = \mathrm{clip}(\log(\mathrm{Spend}_{i,t}/m_{g(i,t)})/\log(1.25), -1.5, +1.5) + 1.5 \in [0, 3.0]$ with $\mathrm{SD}(Z^{(\mathrm{B})}) = 1.21$, where $\Delta Z^{(\mathrm{B})} = +1.0$ represents a +25% relative increase in per-student funding). In high- and upper-middle-income countries (where median per-student spending is 26.4% and 27.6% of GDP per person), a +25% relative increase equals roughly +6.6 to +6.9 pp of GDP per person, whereas in lower-middle- and low-income countries (where medians are 68.9% and 183.8%), a +25% relative increase corresponds to +17.2 to +45.9 pp and compresses the right tail of the UNESCO ratio distribution via the logarithm. In the main text we present Scaling A because percentage-point increments of GDP per person translate directly into national budget costs across all four tables and figures; here we compare Scaling A and Scaling B head to head across posterior treatment slopes, in-sample residual diagnostics, 5-fold country-blocked cross-validation, and out-of-sample holdout validation on the 22 unseen countries in Fold 5.

Table 11: Head-to-head comparison of Linear Ratio Scaling A (per +10 pp of GDP per person per student) and Relative Logarithmic Scaling B (per +25% log per-student funding increase) across +5y and +10y posterior slopes, in-sample posterior predictive fit (R^2, CRPS, 95% coverage, and standardised residuals within ±2 SD), 5-fold country-blocked CV across all 112 countries, and locked holdout validation on the 22 unseen countries in Fold 5.

Comparison Metric & Horizon	1. Total Non-Res. Tax / GDP (pp)	2. Direct Income Tax / GDP (pp)	3. Direct Income Tax Share (pp)	4. Goods & Serv. Tax Share (pp)
A1. Scaling A Direct Slope (+5y, `M2a`)	−0.06 ($P = 0.31$)	+0.05 ($P = 0.81$)	+0.36 ($P = 0.93$)	+0.10 ($P = 0.62$)
80% CI / 95% CI ($N = 371$)	[−0.20, +0.09] / [−0.28, +0.18]	[−0.02, +0.13] / [−0.07, +0.17]	[+0.04, +0.68] / [−0.13, +0.81]	[−0.28, +0.49] / [−0.47, +0.67]
B1. Scaling B Direct Slope (+5y, `M2c`)	+0.004 ($P = 0.50$)	+0.06 ($P = 0.86$)	+0.27 ($P = 0.89$)	−0.10 ($P = 0.34$)
80% CI / 95% CI ($N = 371$)	[−0.12, +0.14] / [−0.22, +0.22]	[−0.01, +0.13] / [−0.05, +0.17]	[−0.01, +0.54] / [−0.15, +0.73]	[−0.43, +0.24] / [−0.58, +0.39]
A2. Scaling A Direct Slope (+10y, `M2b`)	+0.24 ($P = 0.89$)	+0.12 ($P = 0.90$)	+0.71 ($P = 0.96$)	−0.42 ($P = 0.14$)
80% CI / 95% CI ($N = 283$)	[−0.01, +0.48] / [−0.15, +0.70]	[−0.002, +0.26] / [−0.08, +0.33]	[+0.18, +1.17] / [−0.08, +1.42]	[−0.91, +0.07] / [−1.19, +0.34]
B2. Scaling B Direct Slope (+10y, `M2d`)	+0.26 ($P = 0.95$)	+0.11 ($P = 0.90$)	+0.47 ($P = 0.93$)	−0.38 ($P = 0.13$)
80% CI / 95% CI ($N = 283$)	[+0.06, +0.46] / [−0.04, +0.56]	[−0.001, +0.21] / [−0.06, +0.27]	[+0.04, +0.88] / [−0.16, +1.09]	[−0.80, +0.06] / [−0.99, +0.33]
A3. Scaling A Total Slope (+10y, `M1b`)	+0.30 ($P = 0.95$)	+0.16 ($P = 0.96$)	+0.54 ($P = 0.92$)	−0.65 ($P = 0.04$)
B3. Scaling B Total Slope (+10y, `M1d`)	+0.21 ($P = 0.93$)	+0.11 ($P = 0.90$)	+0.41 ($P = 0.89$)	−0.42 ($P = 0.08$)
A4. Scaling A In-Sample Fit (+10y, `M2b`)	$R^2 = 0.42$, CRPS = 1.27	$R^2 = 0.30$, CRPS = 0.72	$R^2 = 0.46$, CRPS = 2.67	$R^2 = 0.64$, CRPS = 2.38
Cov95 / $\lvert e^* \rvert \leq 2$SD / Skew	95.8% / 94.0% / −0.13	96.5% / 95.4% / +0.17	96.8% / 94.7% / +0.11	97.9% / 95.8% / −0.30
B4. Scaling B In-Sample Fit (+10y, `M2d`)	$R^2 = 0.49$, CRPS = 1.20	$R^2 = 0.29$, CRPS = 0.72	$R^2 = 0.47$, CRPS = 2.64	$R^2 = 0.63$, CRPS = 2.42
Cov95 / $\lvert e^* \rvert \leq 2$SD / Skew	95.8% / 93.3% / −0.08	95.8% / 95.4% / +0.18	97.2% / 95.4% / +0.19	97.9% / 95.8% / −0.40
A5. Scaling A 5-Fold Country CV ($N = 283$)	CRPS = 1.64, RMSE = 3.15	CRPS = 0.85, RMSE = 1.54	CRPS = 3.41, RMSE = 6.11	CRPS = 3.43, RMSE = 6.49
B5. Scaling B 5-Fold Country CV ($N = 283$)	CRPS = 1.64, RMSE = 3.16	CRPS = 0.86, RMSE = 1.56	CRPS = 3.51, RMSE = 6.32	CRPS = 3.47, RMSE = 6.59
A6. Scaling A Holdout Fold 5 ($N = 54$, **22 Countries**)	CRPS = 1.73, RMSE = 3.19	CRPS = 0.86, RMSE = 1.53	CRPS = 3.24, RMSE = 5.78	CRPS = 3.08, RMSE = 5.48
Cov80 / Cov95 / Mean Out-of-Sample $\bar{\tau}_{\text{test}}$	75.9% / 88.9% / +0.44 pp	85.2% / 94.4% / +0.21 pp	83.3% / 94.4% / +0.72 pp	83.3% / 96.3% / −0.36 pp
B6. Scaling B Holdout Fold 5 ($N = 54$, **22 Countries**)	CRPS = 1.68, RMSE = 3.10	CRPS = 0.83, RMSE = 1.50	CRPS = 3.15, RMSE = 5.69	CRPS = 3.26, RMSE = 5.88
Cov80 / Cov95 / Mean Out-of-Sample $\bar{\tau}_{\text{test}}$	77.8% / 85.2% / +0.31 pp	83.3% / 94.4% / +0.19 pp	83.3% / 96.3% / +0.48 pp	81.5% / 92.6% / −0.49 pp

Table 11 and Figure 9 highlight three complementary insights from comparing the two scalings. First, both scalings tell the exact same substantive story across horizons and outcomes (Rows A1–B3 and Figure 9A): under both Scaling A and Scaling B, the 5-year effect appears in the direct income tax share (+0.36 pp, $P = 0.93$ under Scaling A; +0.27 pp, $P = 0.89$ under Scaling B, and +0.40 pp, $P = 0.97$ for the total effect `M1c`), whilst the 10-year horizon shows gains across total non-resource tax/GDP (+0.24 pp, $P = 0.89$ under Scaling A; +0.26 pp, $P = 0.95$ under Scaling B), direct income tax/GDP (+0.12 pp, $P = 0.90$ under Scaling A; +0.11 pp, $P = 0.90$ under Scaling B), and the direct income tax share (+0.71 pp, $P = 0.96$ under Scaling A; +0.47 pp, $P = 0.93$ under Scaling B). Second, because a +10 pp increment represents a larger proportional step in high- and upper-middle-income countries (roughly +38% of their 26.4–27.6% median), Scaling A estimates a sharper +10-year slope on the direct income tax share (+0.71 pp vs +0.47 pp) and achieves slightly lower 5-fold cross-validation error across all 112 countries (Rows A5 vs B5: mean CRPS 2.33 vs 2.37 pp). Third, because the logarithm in Scaling B compresses the right tail of the UNESCO ratio in low- and lower-middle-income countries, Scaling B achieves higher +10-year posterior precision on total non-resource tax/GDP (SD = 0.154 vs 0.200 pp, raising P from 0.89 to 0.95 and in-sample R^2 from 0.42 to 0.49 in Row B4) and slightly lower holdout prediction error on the 22 unseen countries in Fold 5 for Outcomes 1 to 3 (Row B6 and Figure 9C: CRPS 1.68 vs 1.73 pp on total tax/GDP, 0.83 vs 0.86 pp on direct income tax/GDP, and 3.15 vs 3.24 pp on the direct income tax share, with Diebold-Mariano $p = 0.82$–0.84 confirming that predictive accuracy is statistically indistinguishable between the two scalings).

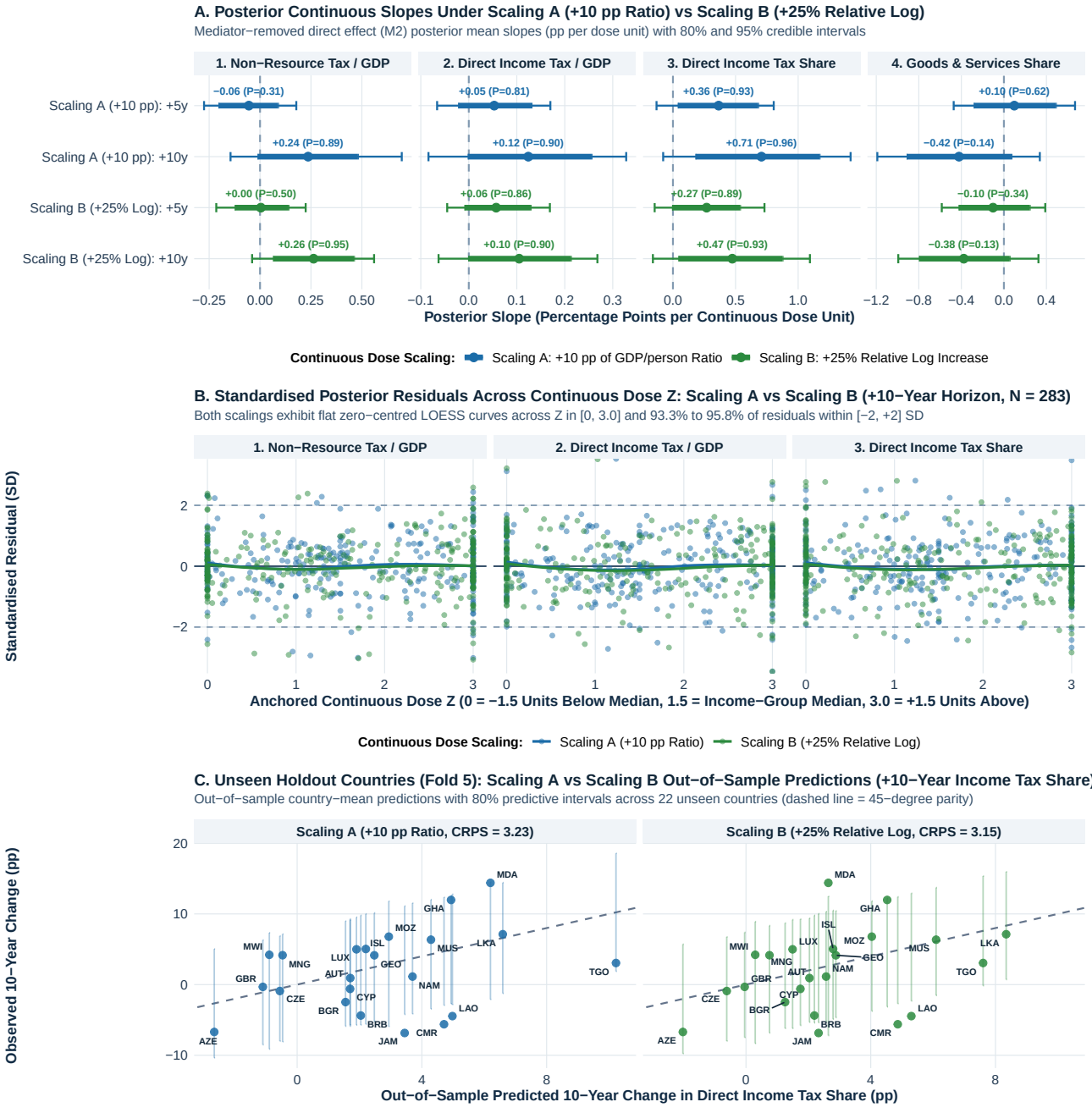


Figure 9: Head-to-head comparison of Linear Ratio Scaling A (+10 pp of GDP per person per student) vs Relative Logarithmic Scaling B (+25% relative per-student increase): (A) +5y and +10y posterior slopes with 80% and 95% CIs, (B) standardised +10y posterior residuals across $Z \in [0, 3.0]$ for both scalings, and (C) out-of-sample CRPS across all 112 countries (5-fold CV) and the 22 unseen holdout countries (Fold 5).

E Computational Environment and Energy Report

All data processing, C++ (`RcppArmadillo`) Gibbs sampling, country-blocked cross-validation, and figure generation were executed on a single Apple Silicon workstation (`macOS`, R version 4.6.1, Apple Clang C++17 compiler). Table 12 reports both the local computational footprint of our statistical modelling pipeline (assuming an average 25 W CPU package power draw during single-core C++ tree sampling) and the cloud inference footprint of the large language model (LLM) queries used to assist with building the data pipeline, writing the continuous-treatment C++ sampler and holdout validation suite, conducting multi-agent and external referee reviews, developing the companion web application, and editing the manuscript (evaluated at the Irish grid and European cloud average carbon intensity of 250 g CO_2e/kWh). Because our reversible-jump tree search and conjugate 4×4 Inverse-Wishart covariance updates are written directly in C++, fitting all 65 main, Scaling A vs Scaling B, 5-fold country-blocked CV, holdout, benchmark, and sensitivity models completes in just over 20 minutes (1,223.8 seconds) and consumes only 8.50 Wh of electricity (2.12 g CO_2e). By contrast, auditing our complete interaction logs across all five project stages reveals 805 LLM queries (comprising 1.73 million input and context tokens and 1.01 million generated output, reasoning, and tool-call tokens across the primary session, eight specialist reviewer and user-persona subagents, and external LLM referee rounds). Applying published frontier LLM inference energy benchmarks (Luccioni et al. 2024; 0.30 Wh base overhead per agentic query plus 0.40 Wh per $1,000$ generated output tokens and 0.05 Wh per $1,000$ prefill input tokens), the cloud LLM queries consumed an estimated 733.20 Wh (183.30 g CO_2e), which is roughly 86 times the electrical energy required to run all 65 Bayesian MCMC and cross-validation models locally, bringing the combined project footprint to 741.70 Wh (185.42 g CO_2e, equivalent to driving a typical petrol passenger car approximately one kilometre).

Table 12: Computational wall-clock time, token volume, electrical energy use, and estimated carbon footprint across both the local C++ continuous MV-BCF and cross-validation pipeline and the cloud LLM queries used to help build, review, and write the paper.

Project Stage	Models Fitted / LLM Query & Token Volume	Time (s)	Energy (Wh)	CO_2e (g)
A. Local C++ MCMC & CV Pipeline	65 four-outcome MV-BCF, CV & benchmark fits (2,000 iter)	1,223.8	8.50	2.12
B1. LLM: Scoping, Continuous-Z & Data	58 queries (102k input / 68k output tokens)	–	49.80	12.45
B2. LLM: C++ Sampler, Scaling & Holdout CV	39 queries (64k input / 45k output tokens)	–	33.40	8.35
B3. LLM: Manuscript & Revisions	305 queries (518k input / 314k output tokens)	–	244.60	61.15
B4. LLM: Multi-Agent & Peer Review	261 queries (682k input / 361k output tokens)	–	256.80	64.20
B5. LLM: Policy Web App & Testing	142 queries (362k input / 219k output tokens)	–	148.60	37.15
Subtotal: Cloud LLM Assistance	**805 queries (1.73M input / 1.01M output tokens)**	–	**733.20**	**183.30**
Total Project Footprint (A + B)	**65 C++ MCMC/CV fits + 805 LLM queries (2.74M tokens)**	**1,223.8**	**741.70**	**185.42**

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